Computer Vision and Machine Learning for Scientific Visual Data Analytics

Lecture 7: Advanced Topics

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OIST

Roadmap

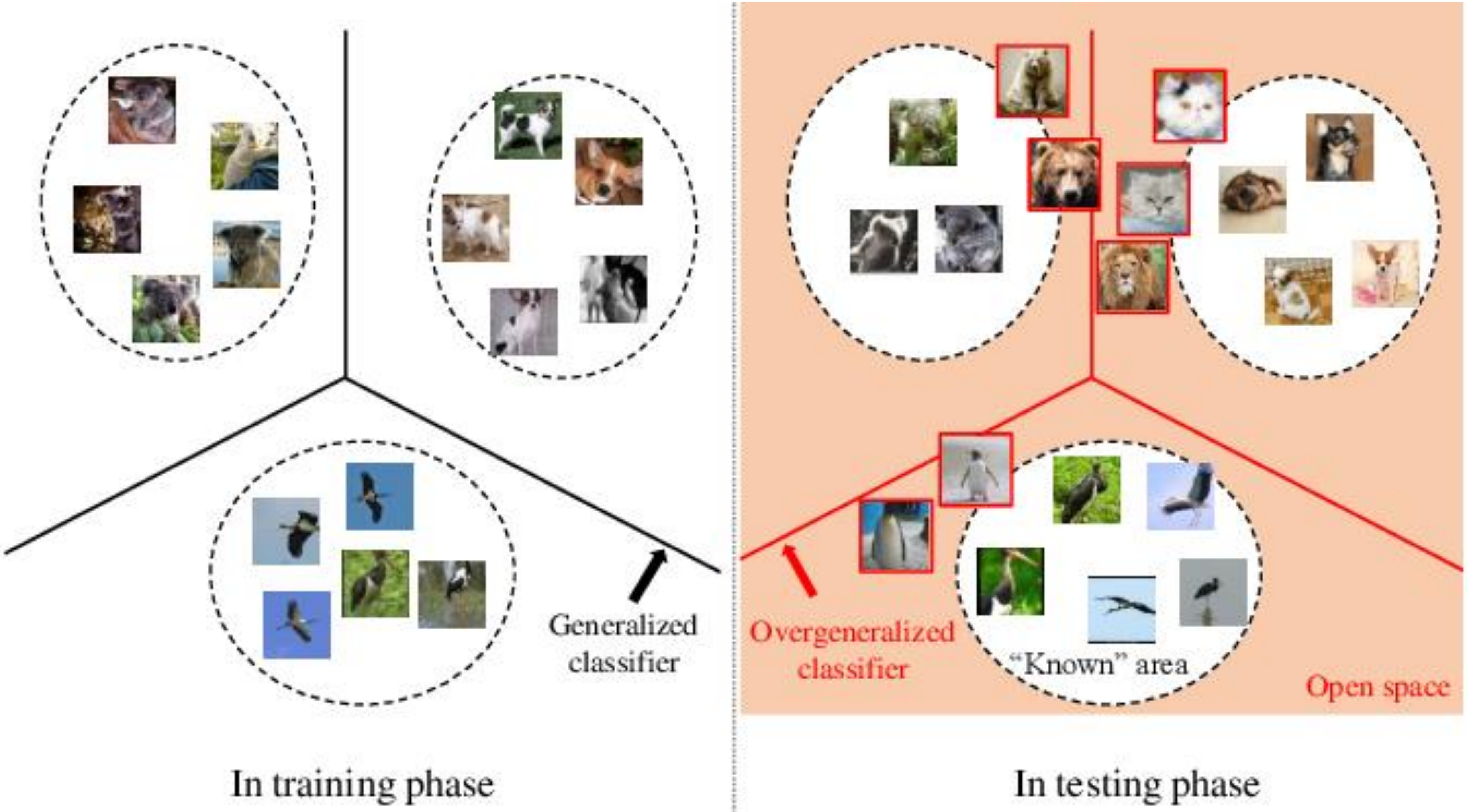
Teaching materials are available: https://github.com/aimerykong/OIST-mini-course-CVML

Lecture	Topics	Hands-on experience
1. Intro	CV/ML, deep learning, artificial intelligence	Configure programming environment using Google Drive and Colab; implement basic scripts in Python
2. Basics of CV/ML	training, validation, testing, features, metrics, nearest neighbor	Implement a machine learning pipeline for image classification
3. Classification	Logistic Regression, gradient descent	Train a linear classifier for pollen species recognition
4. Deep Learning	Multi-Layer Perceptron, feature learning, backpropagation, neural networks	Train a convolutional neural network (CNN) for pollen species recognition
5. Segmentation and Detection	semantic and instance segmentation, U-Net,	Train a neural network for pollen grain detection/segmentation
6. Foundation Models	CLIP, Segment Anything Model, DINOv2, GroundingDINO, finetuning	Finetune a pretrained model for image classification and segmentation
7. Advanced Topics	open-set recognition, novel species recognition, phylogentic reconstruction via clustering	Implement algorithms to recognize unknown/anomalous/novel observations
8. Brainstorm & Talks	Volunteer to present	n/a

Outline

- Open-Set Recognition for Novel Species Detection
- Clustering for Phylogenetic Reconstruction

Open-set vs. closed-set



Open-set: some species you never saw before

Open-set comes in different ways.

- Novel species

Open-set comes in different ways.

- Novel poses and modes

Tesla Model 3 crashing into an overturned truck

Thought experiment: How do you decide something is unknown to you?

Suppose you are required to recognize an object in an image,

- When do you say “I don’t know”?
- How do you decide “I don’t know”?

Thought experiment: How do you decide something is unknown to you?

Suppose you are required to recognize an object in an image,

- When do you say "I don't know"?
- How do you decide "I don't know"?

Something you are not ***familiar*** with!
How to measure "***familiar***"?
Not like what you know...?
Compare to what you know...

Thought experiment: How do you decide something is unknown to you?

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Open-set using nearest neighbor

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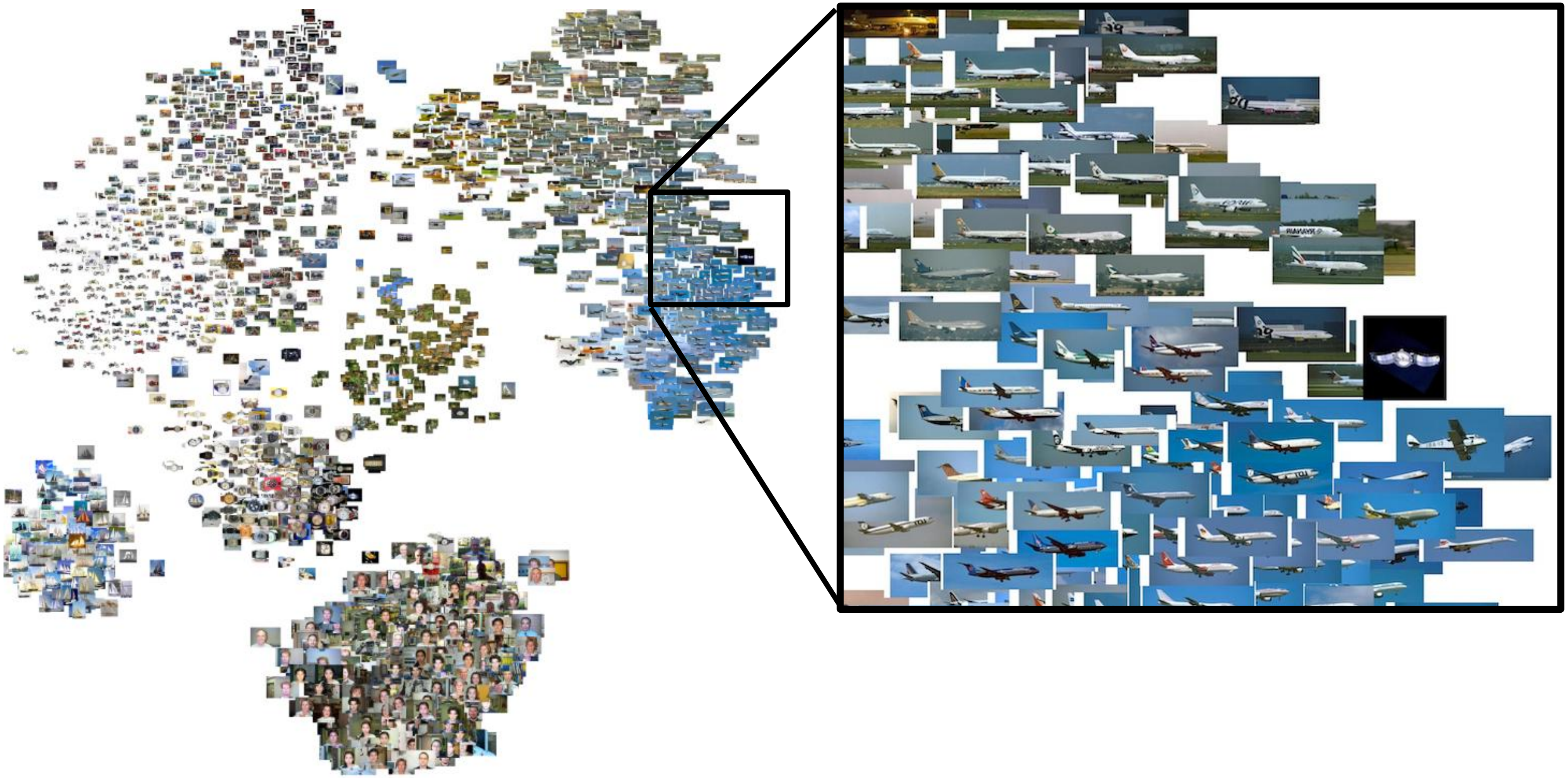
Intuition: if a testing example if far from any seen/training examples, it is more likely to be an open-set/anomalous example.

Open-set using nearest neighbor

Visualization in a feature space



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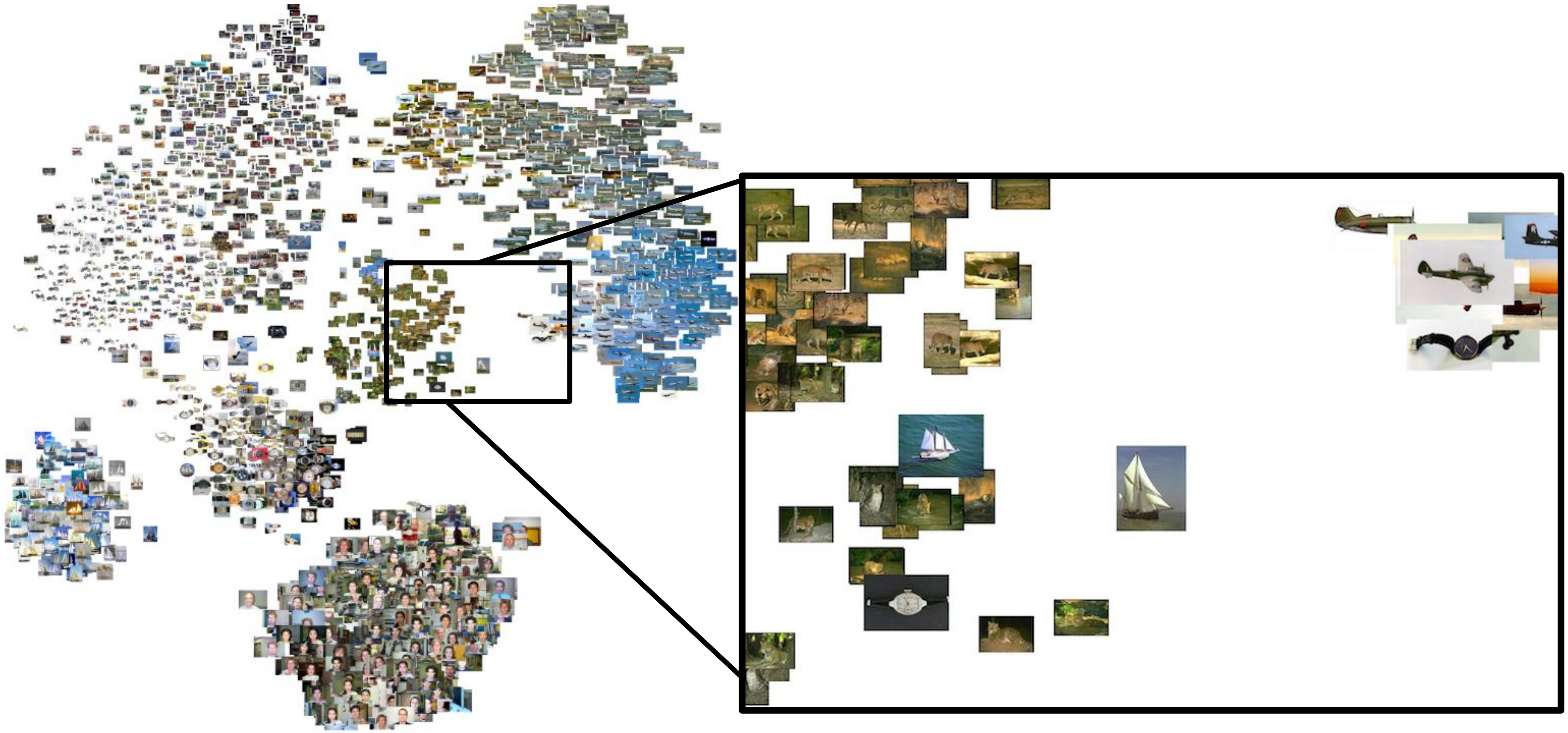

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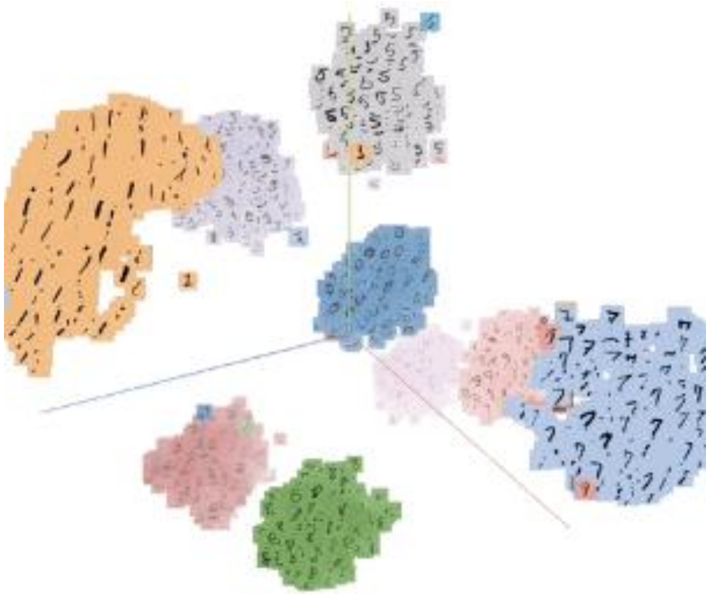
Open-set using nearest neighbor

It is costly to save all the training data.

We human being abstract knowledge instead of memorizing everything we encounter.

Intuition: if a testing example if far from any seen/training examples, it is more likely to be an open-set/anomalous example.

Open-set using Nearest Class/Cluster Mean (NCM)



Let's abstract knowledge for the data!

e.g., cluster data and represent them using the cluster mean (plus deviation).

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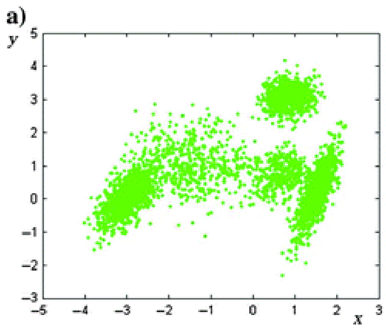

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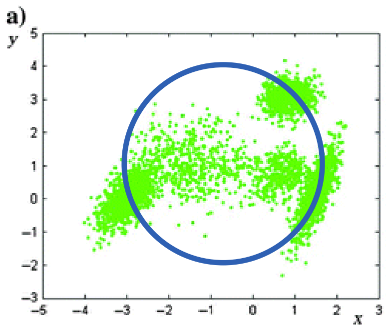

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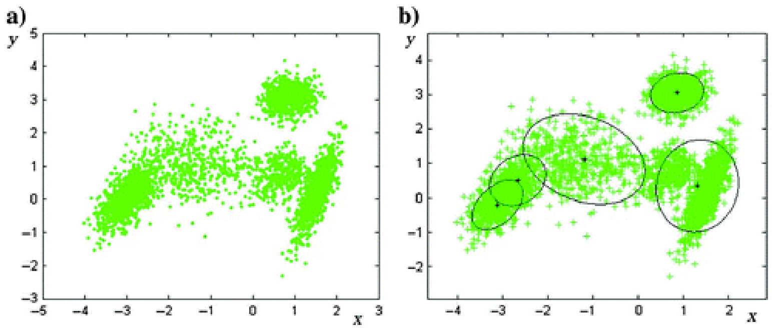

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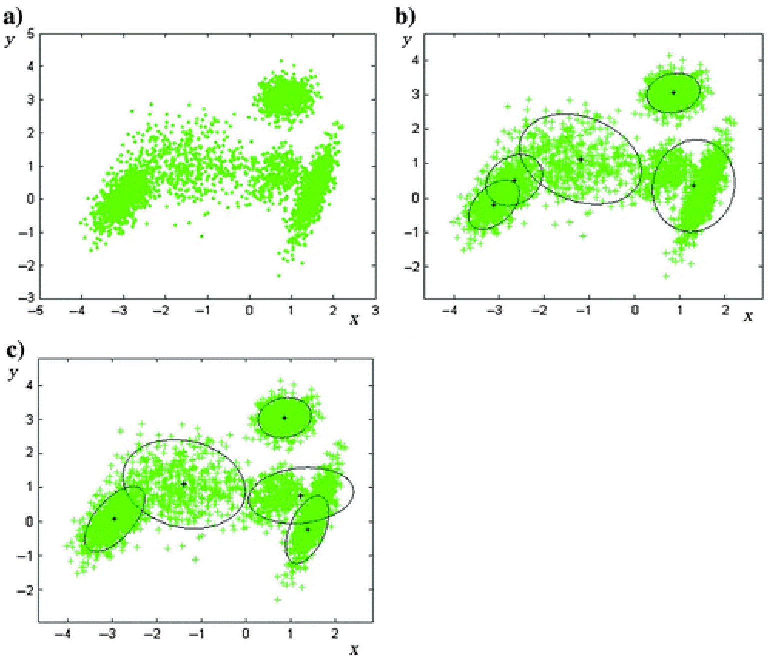

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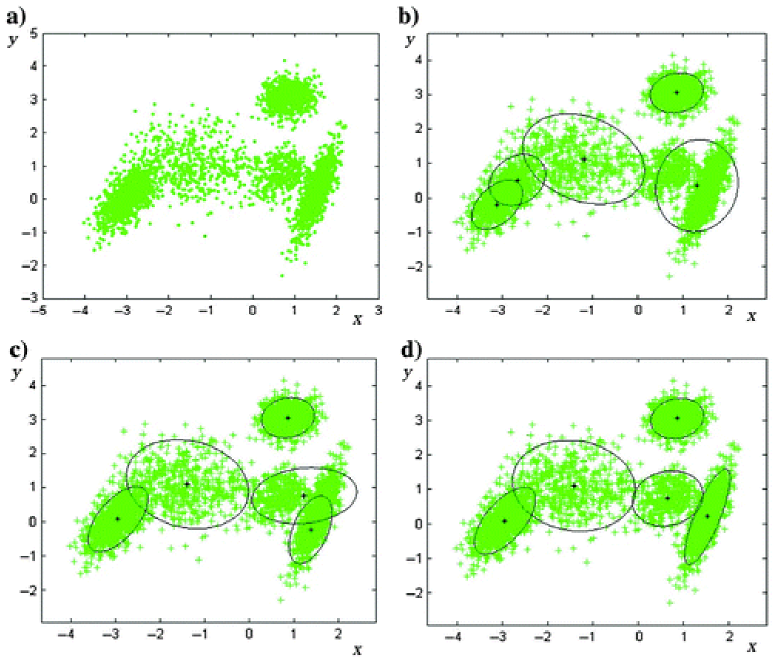

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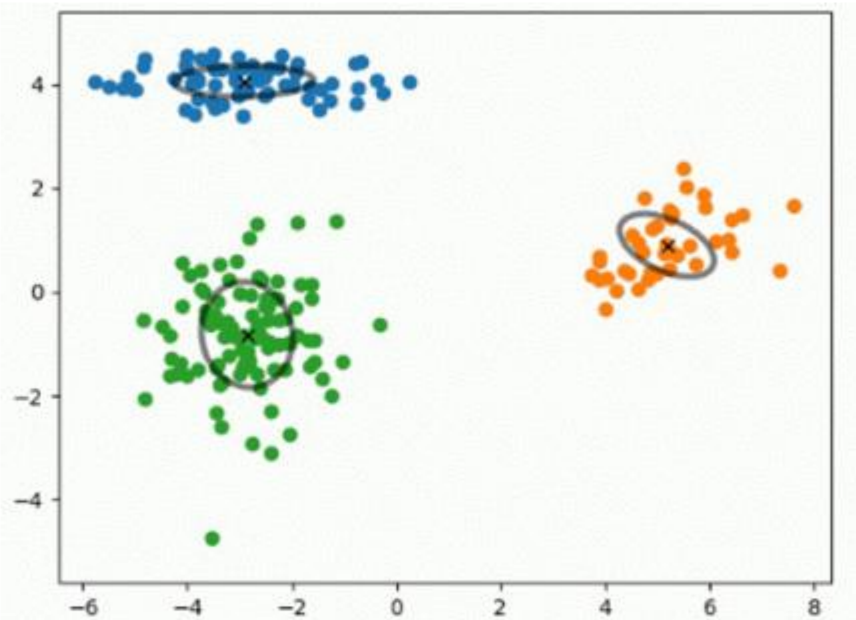

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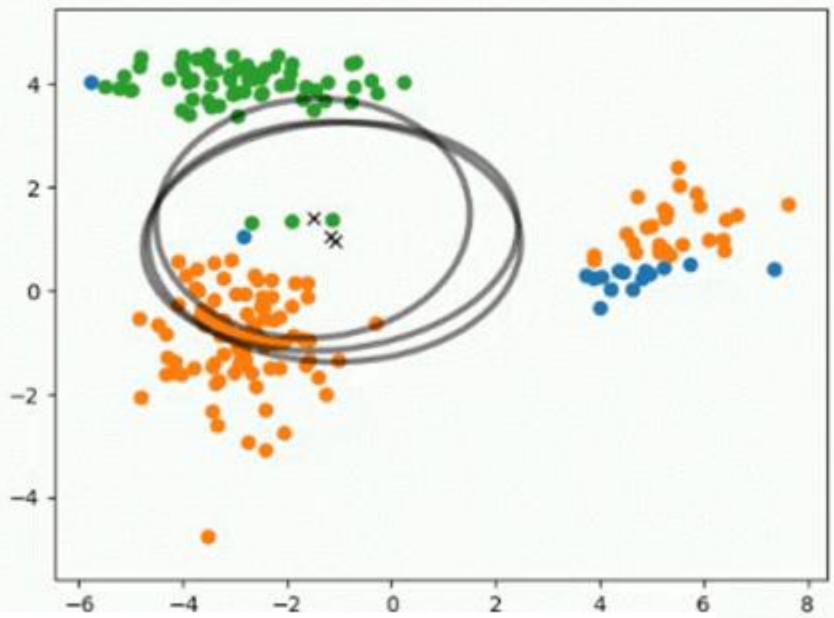
Open-set using Gaussian Mixture Model



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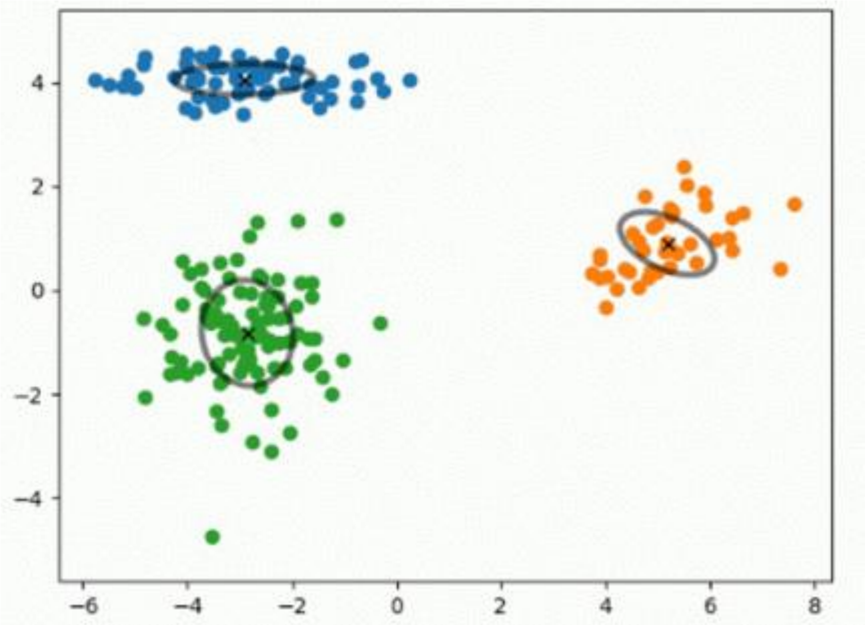

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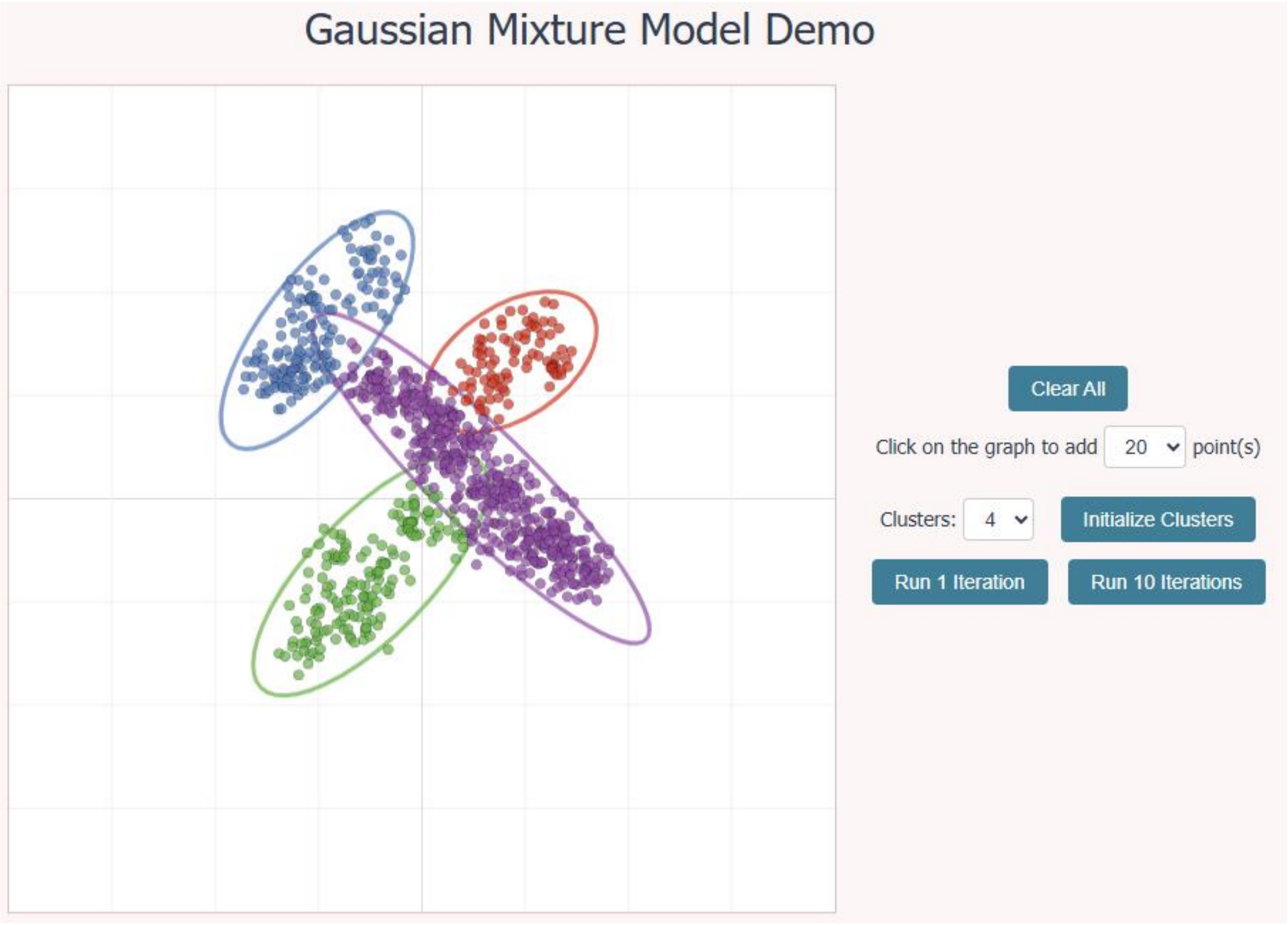
Open-set likelihood will be the inverse of "belongingness" to a closest Gaussian Model

Intuition: if a testing example if far from any **clusters** of seen/training examples, it is more likely to be an open-set/anomalous example.

Demo of Gaussian Mixture Model

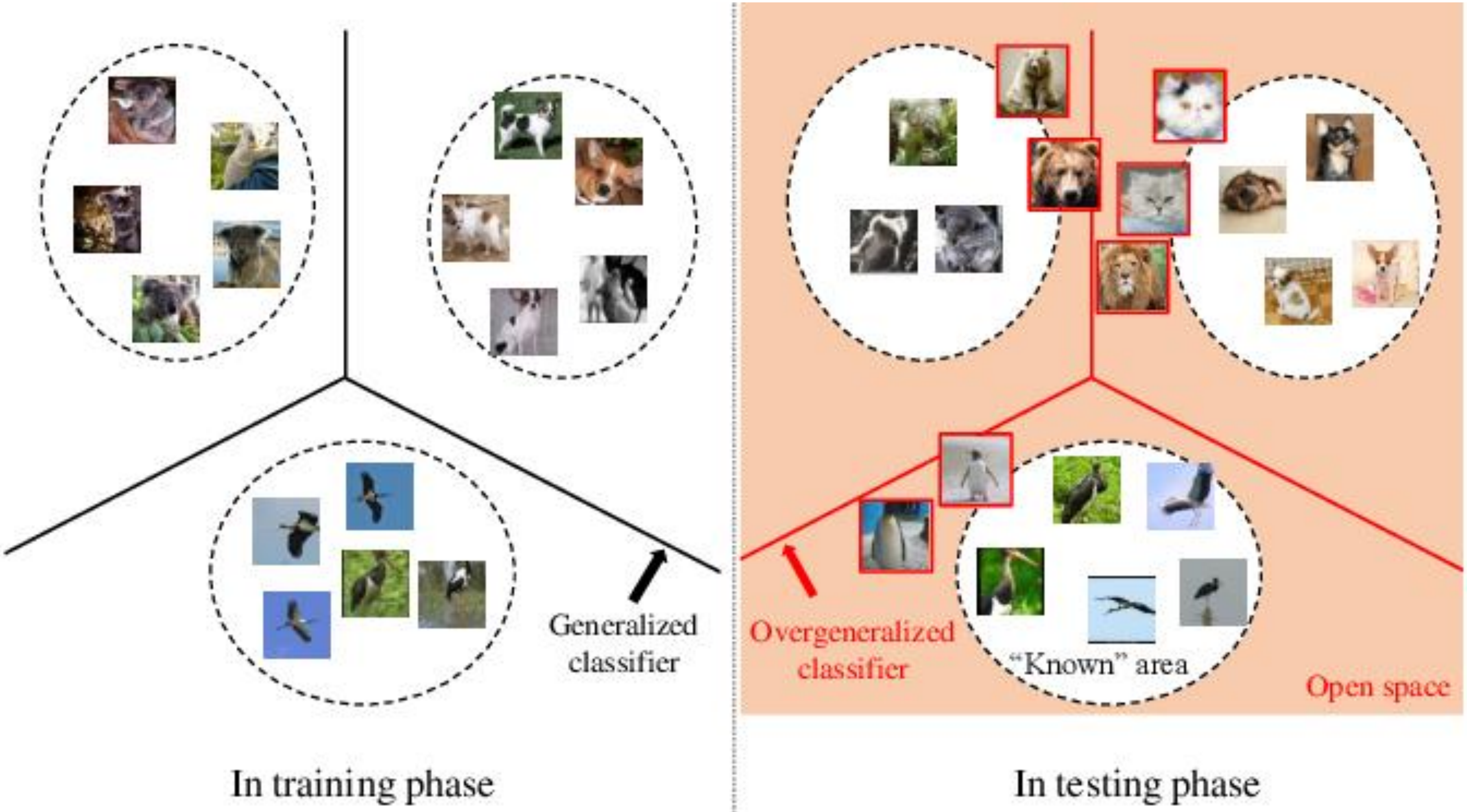
https://lukapopijac.github.io/gaussian-mixture-model



Approaches to open-set recognition

- Generative approach: NN & Nearest Class / Cluster Mean (NCM) / GMM
- **Max of softmax score & max of logit score**
- Entropy
- Outlier exposure
- Open data generation
- Joint force with OpenGAN

Open-set recognition using the softmax classifier

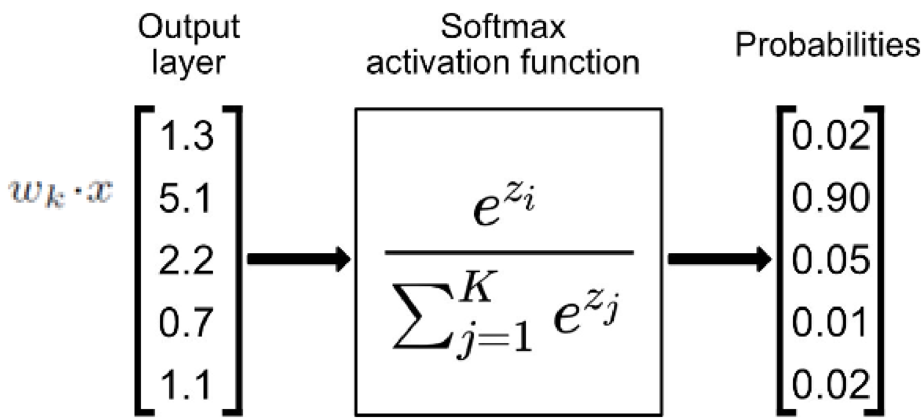


Open-set: data with uncertain predictions, e.g., around decision boundary

Open-set likelihood measured by the max of softmax

Open-set likelihood ← inverse(max$_k$ $p(y=k|x)$)

$$p(y=k|x) = \frac{e^{w_k \cdot x}}{\sum_i e^{w_i \cdot x}}$$



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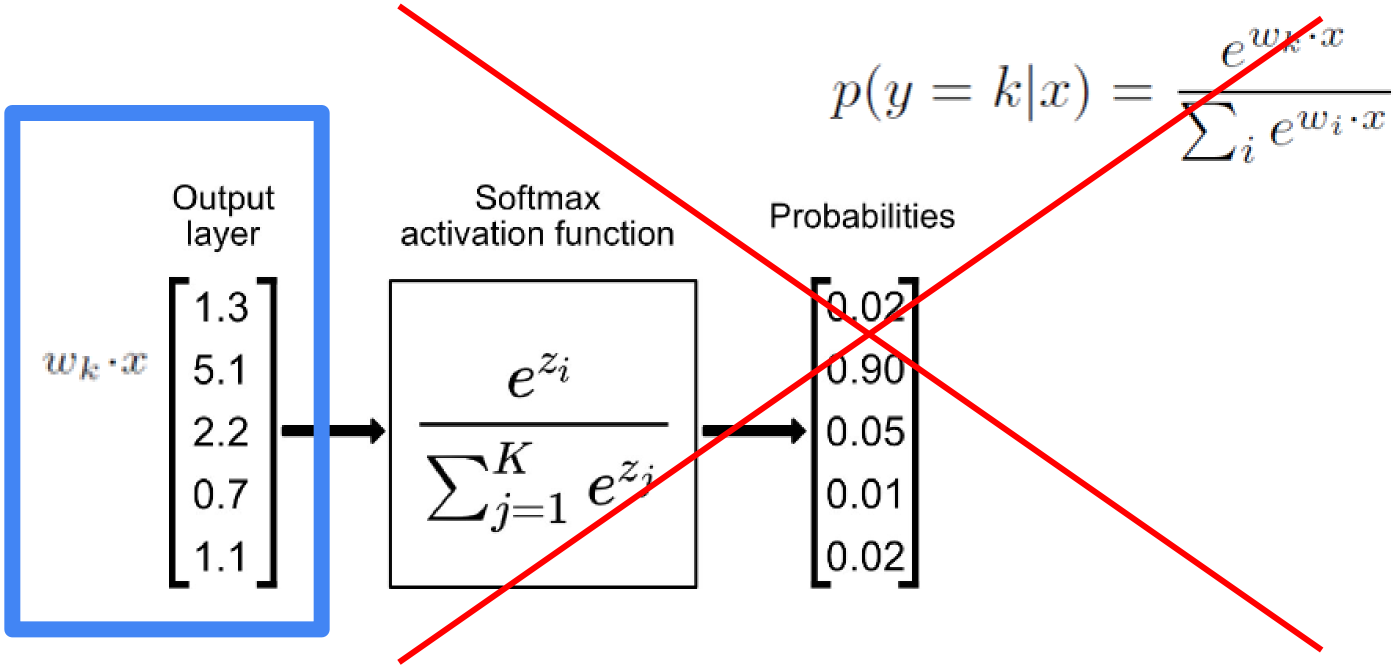
D Hendrycks, K Gimpel, "A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks", ICLR, 2017

Open-set likelihood measured by the max of **logits**

Open-set likelihood <= -inverse(max $w_k^T x$)



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Sagar Vaze, Kai Han, Andrea Vedaldi, Andrew Zisserman, "Open-Set Recognition: a Good Closed-Set Classifier is All You Need?", ICLR, 2022

Approaches to open-set recognition

- Nearest Neighbor & Nearest Class/Cluster Mean (NCM)
- Max of softmax score & max of logit score
- **Entropy**
- Outlier exposure
- Open data generation
- Joint force with OpenGAN

Open-set likelihood measured by entropy

Dictionary

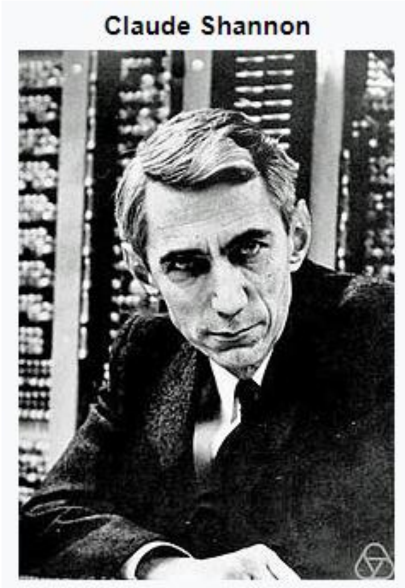
Definitions from Oxford Languages · Learn more

en·tro·py

/ˈentrəpē/

noun

1. PHYSICS
   a thermodynamic quantity representing the unavailability of a system's thermal energy for conversion into mechanical work, often interpreted as the degree of disorder or randomness in the system.
   "the second law of thermodynamics says that entropy always increases with time"
2. lack of order or predictability; gradual decline into disorder.
   "a marketplace where entropy reigns supreme"



Open-set: data with "more disordered / random" predictions

Open-set likelihood measured by entropy

Claude Shannon

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Output layer | Softmax activation function | Probabilities

$$w_k \cdot x \quad \begin{bmatrix} 1.3 \\ 5.1 \\ 2.2 \\ 0.7 \\ 1.1 \end{bmatrix} \longrightarrow \boxed{\frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}} \longrightarrow \begin{bmatrix} 0.02 \\ 0.90 \\ 0.05 \\ 0.01 \\ 0.02 \end{bmatrix}$$

$$p(y=k|x) = \frac{e^{w_k \cdot x}}{\sum_i e^{w_i \cdot x}}$$

$$\boxed{H(p) = -\sum_{k=1}^{C} p(y=k|x) \log(p(y=k|x))}$$

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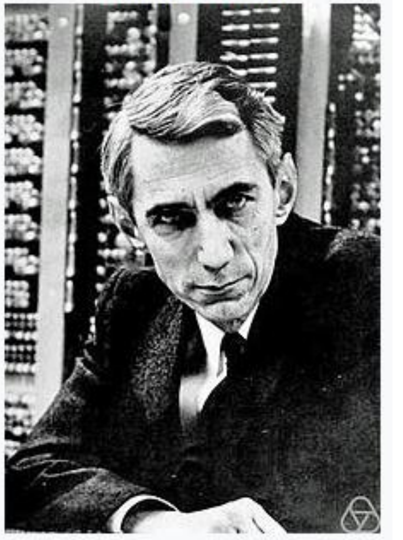

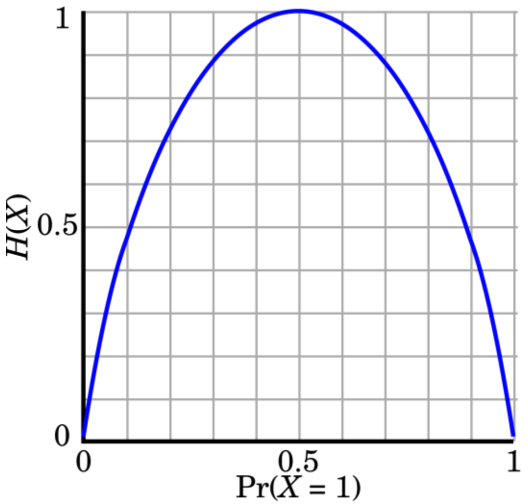

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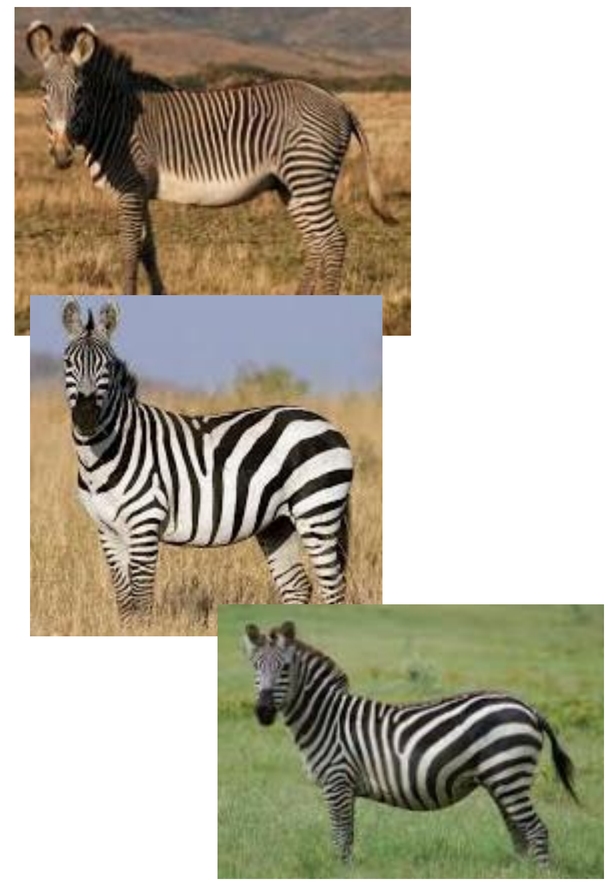
Open-set: data with “more disordered / random” predictions

Open-set likelihood measured by entropy

- Now, you only see kangaroo and zebra, you don't know other things.
- Open-set is anything that are neither a kangaroo nor a zebra.
- Open-set likelihood is a measure of a degree you think the input image is "not a kangaroo and not a zebra".

kangaroo vs. zebra



Open-set likelihood measured by entropy

kangaroo : 0.95
zebra : 0.05
entropy = 0.086

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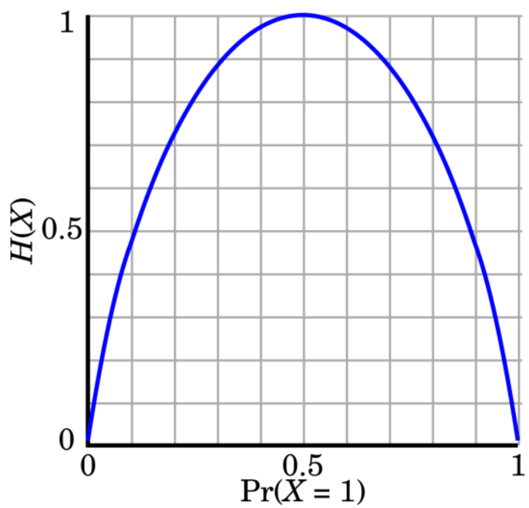

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kangaroo: 0.05
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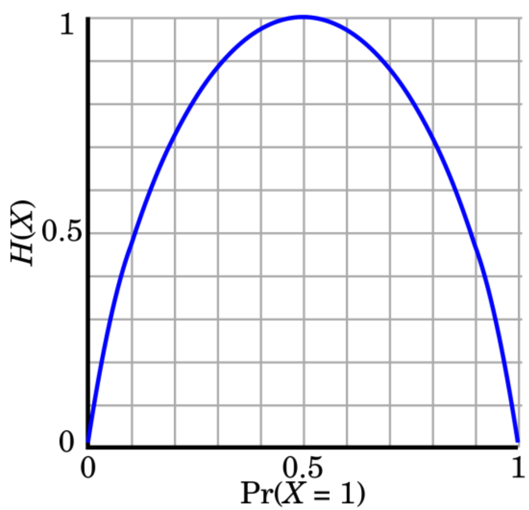
entropy = 0.086

kangaroo: 0.50
zebra : 0.50
entropy = 0.30



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Approaches to open-set recognition

- Nearest Neighbor & Nearest Class/Cluster Mean (NCM)
- Max of softmax score & max of logit score
- Entropy
- **Outlier exposure**
- Open data generation
- Joint force with OpenGAN

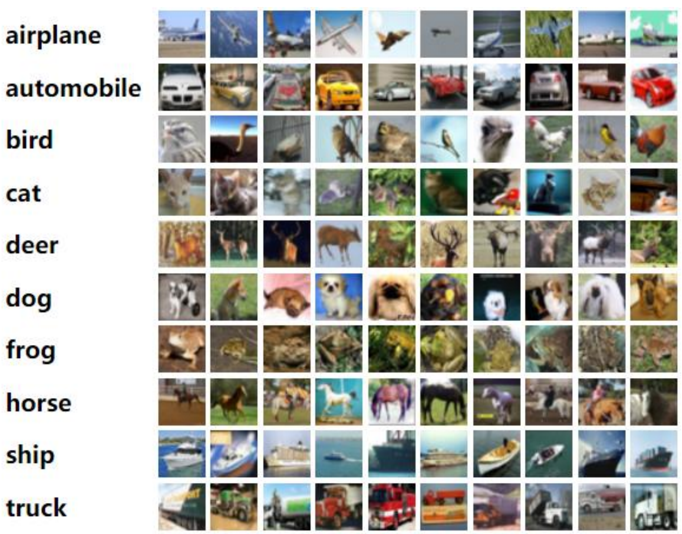
Rethink the nature of open-set recognition

- Recognizing open-set vs. closed-set is essentially a binary classification problem.

Open-set recognition by learning with (pseudo) “open-set” training data

- Collect out-of-domain / outlier data as the pseudo open-set data
- Train a binary classifier to recognize open-set vs. closed set

closed-set training data



Pseudo open training data

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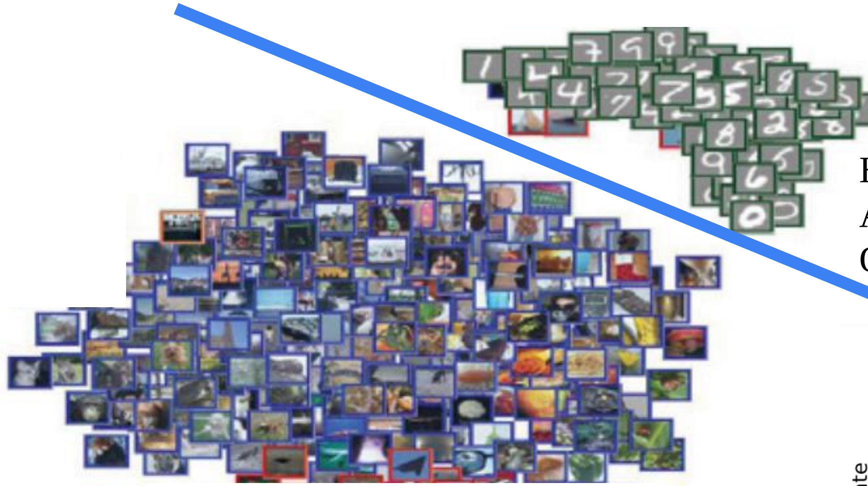
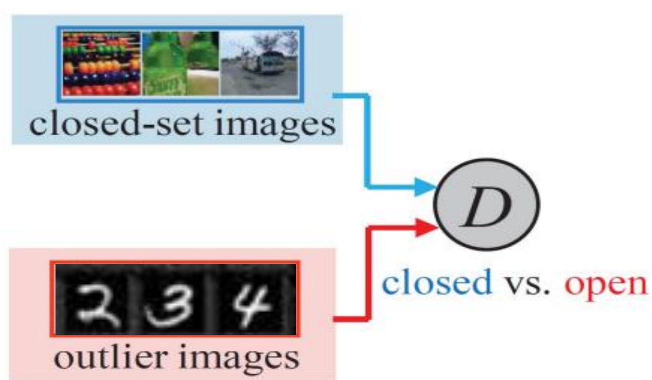




Hendrycks, et al., "Deep Anomaly Detection with Outlier Exposure", ICLR, 2019

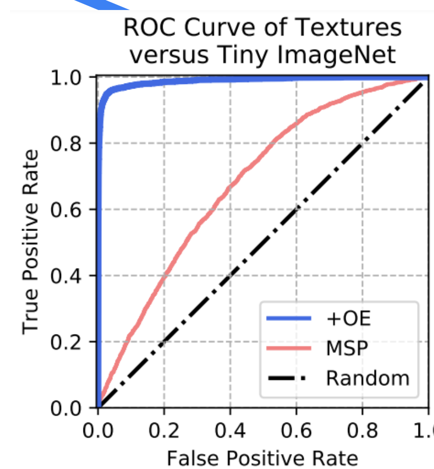
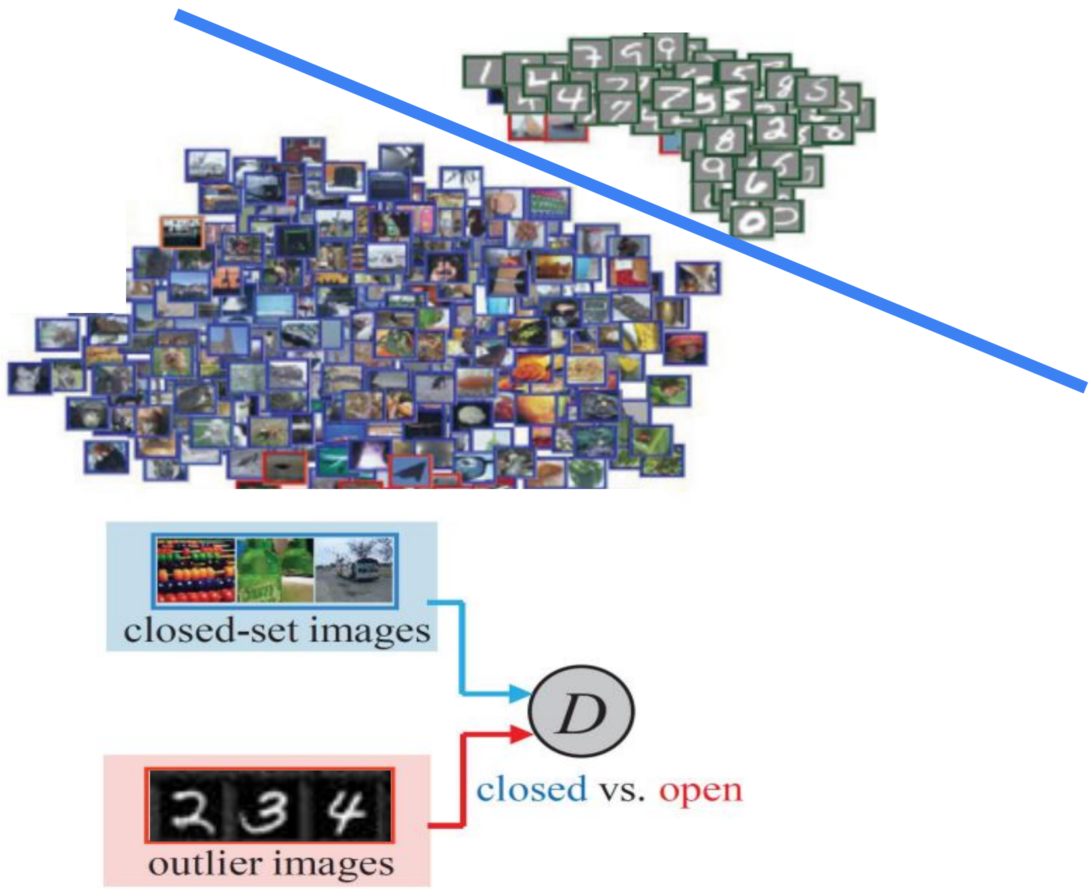


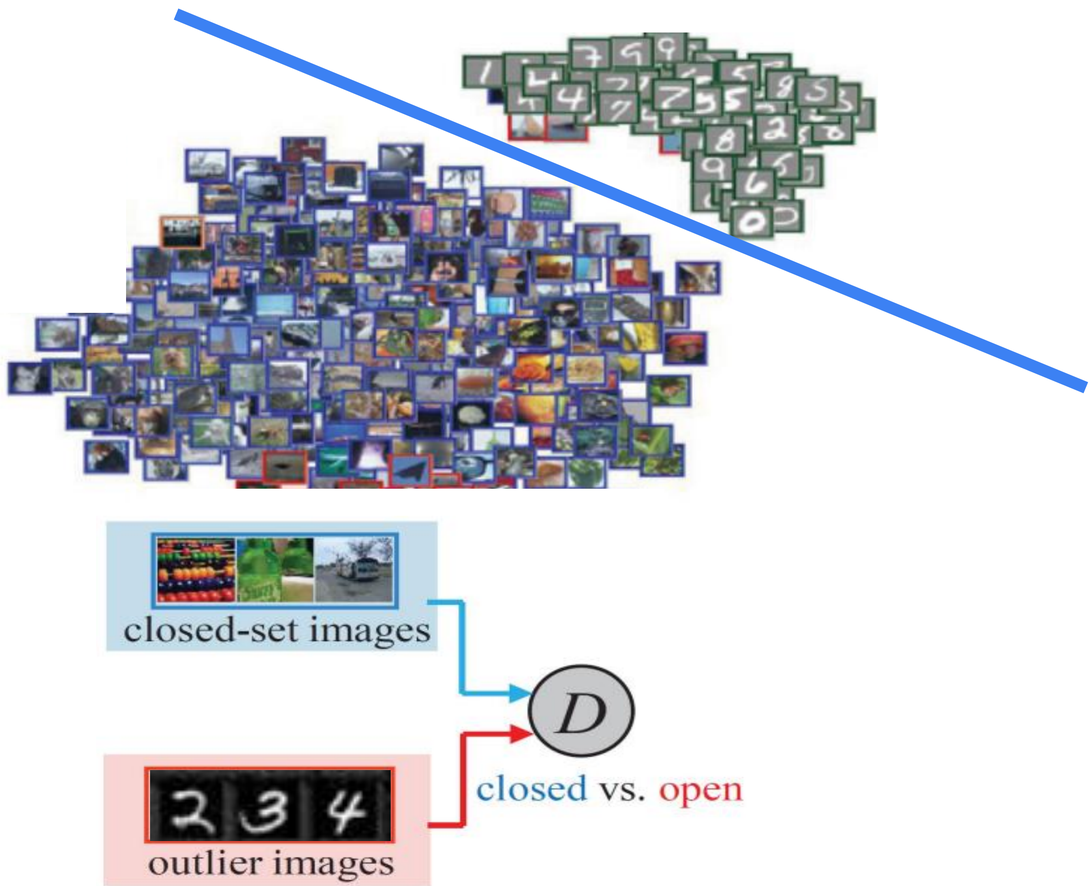
Figure 1: ROC curve with Tiny ImageNet ($\mathcal{D}_{\text{in}}$) and Textures ($\mathcal{D}_{\text{out}}^{\text{test}}$).

Limitation of (pseudo) “open-set” training data



Limitation of (pseudo) "open-set" training data

- They cannot span the whole open world!
- Discriminative open-vs-closed classifiers can overfit, e.g., memorizing the pseudo open-set data and generalizing to real ones poorly.



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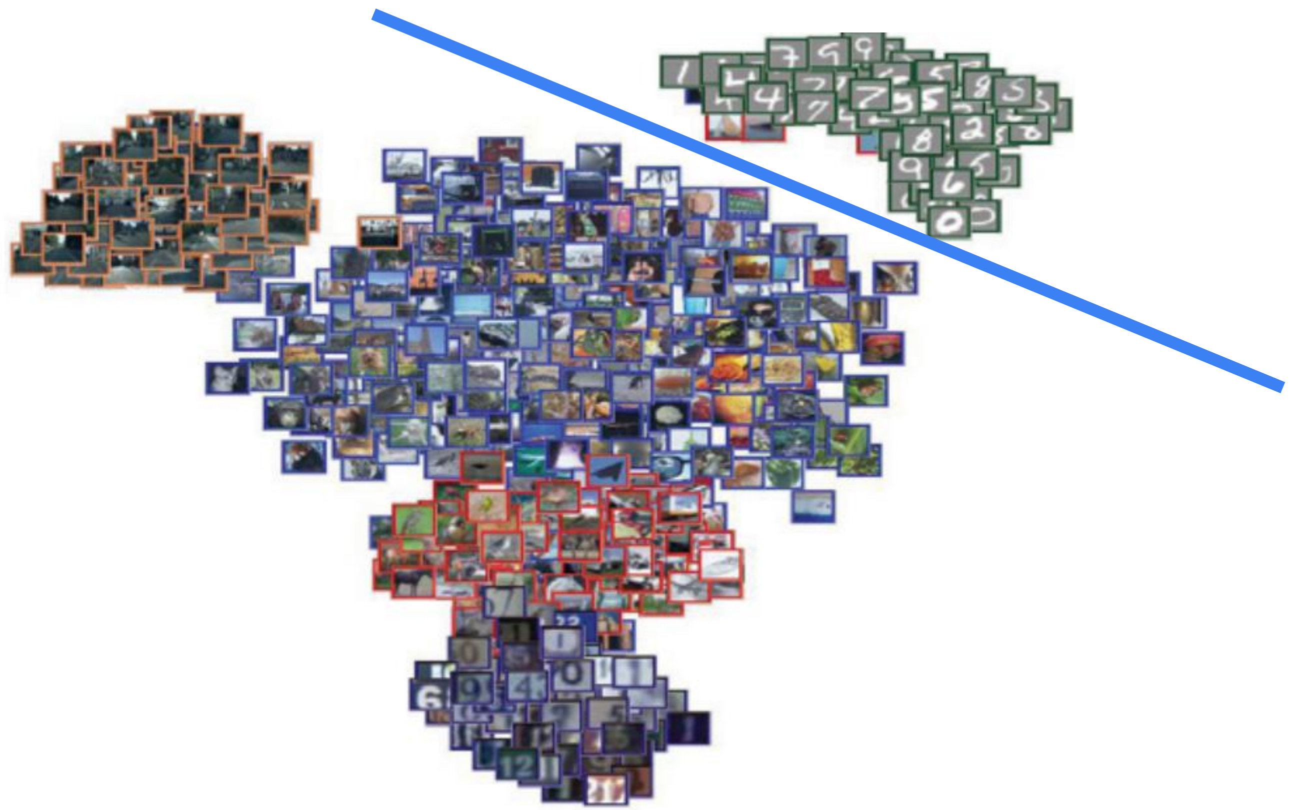

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- Can we find better “pseudo open-set training data”?

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- Can we find better “pseudo open-set training data”?
- How about synthesizing open-set training data?

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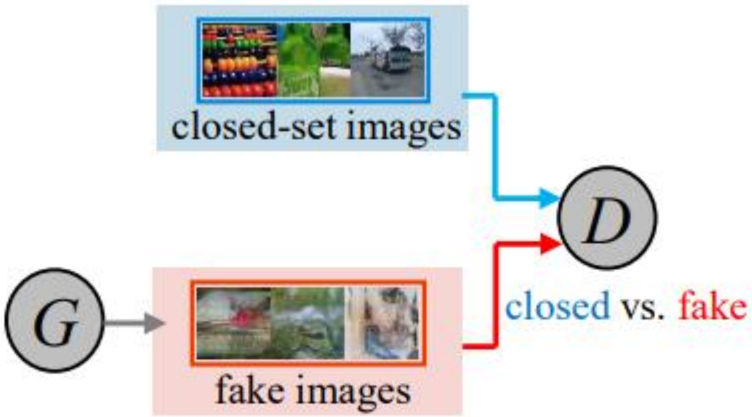
Learn to synthesize the “right” pseudo open-set data

- Hopefully, we have a smart generator to synthesize such data for us!



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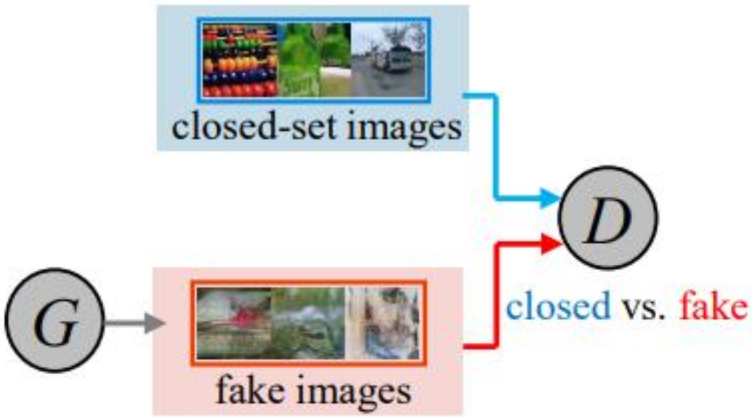

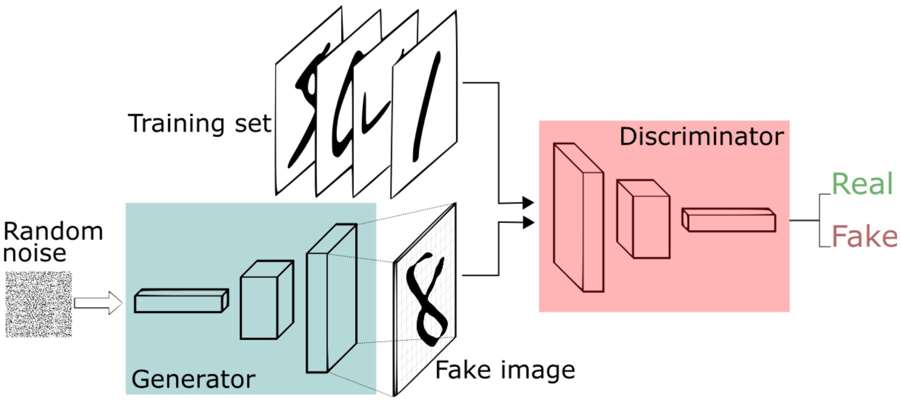

Goodfellow, et al., "Generative Adversarial Networks:, NeurIPS, 2014

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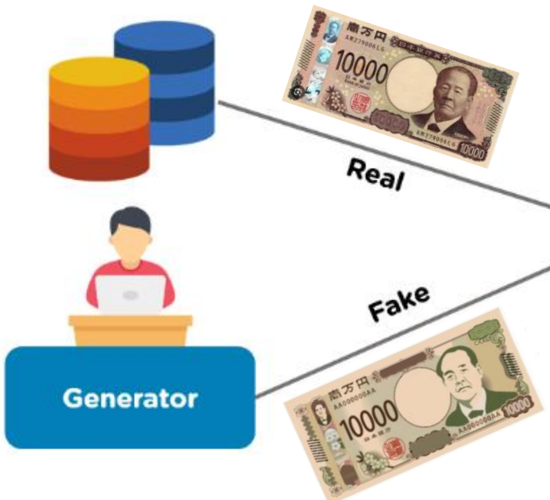

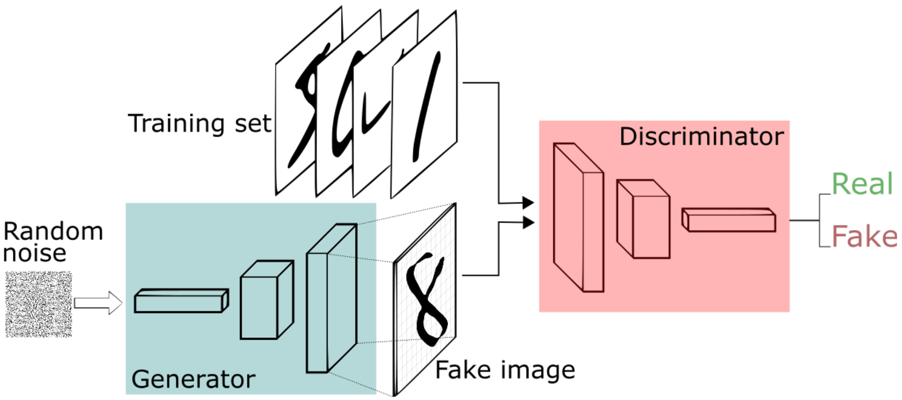

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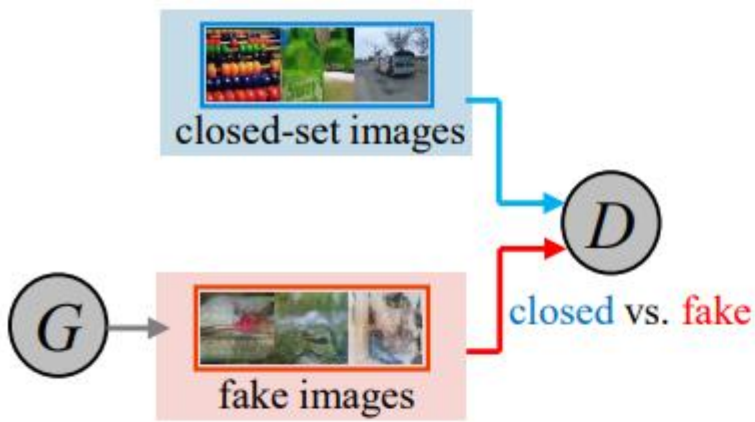

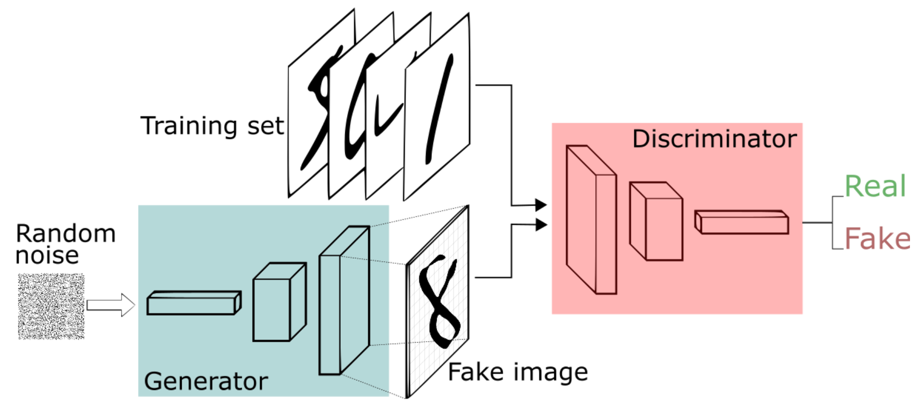

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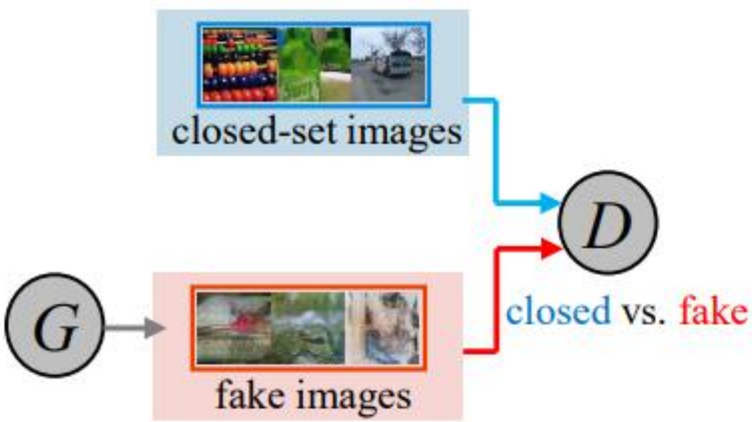
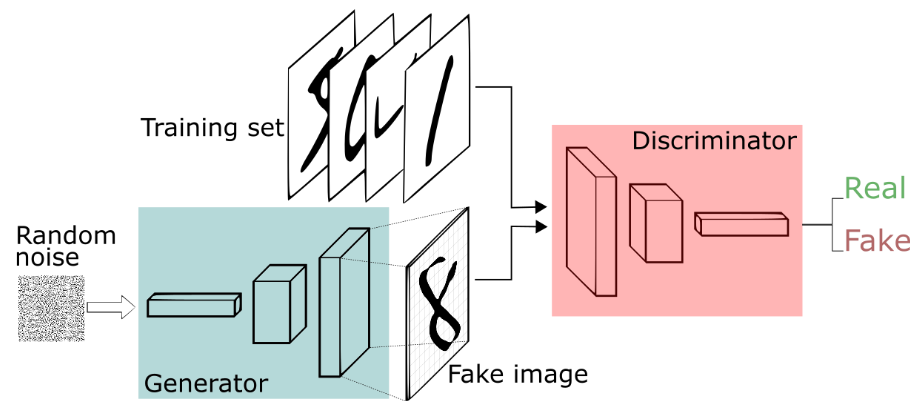
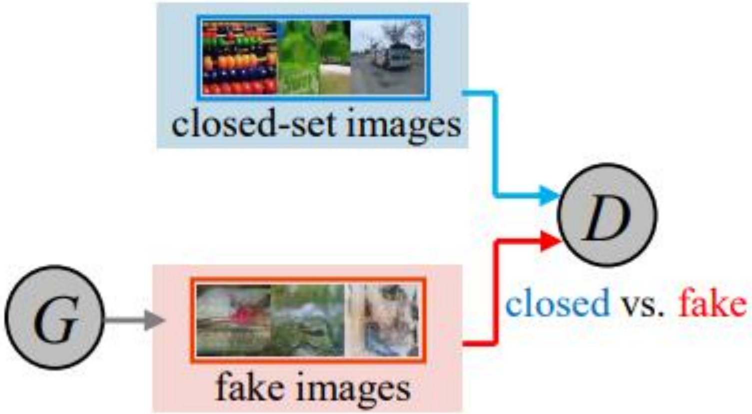
- Generative Adversarial Network (GAN)
- Importantly, the real-vs-fake classifier can be used as closed-vs-open classifier!

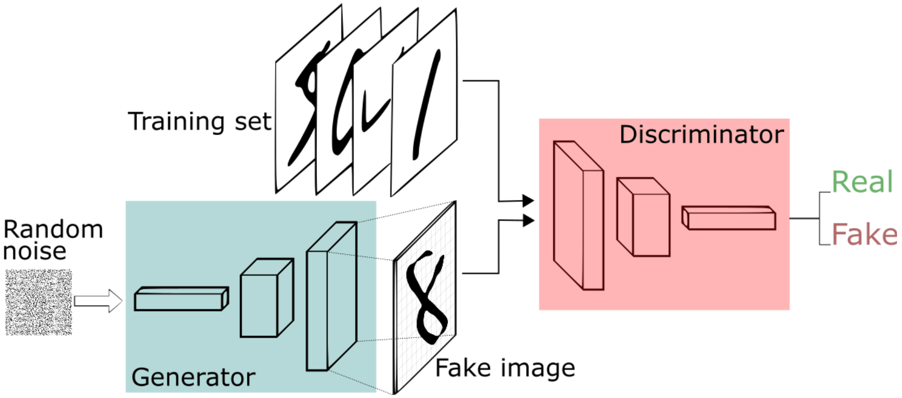


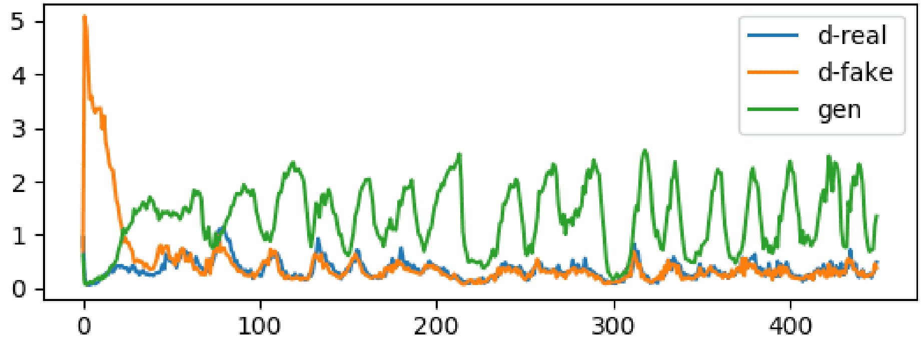


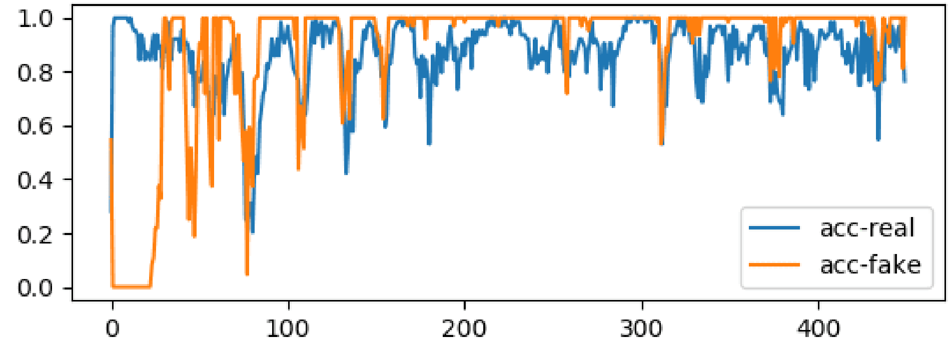
Learn to synthesize the "right" pseudo open-set data

- Hopefully, we have a smart generator to synthesize such data for us!
- Generative Adversarial Network (GAN)
- Importantly, the real-vs-fake classifier can be used as closed-vs-open classifier!
- However, adversarial/alternative training is not stable.







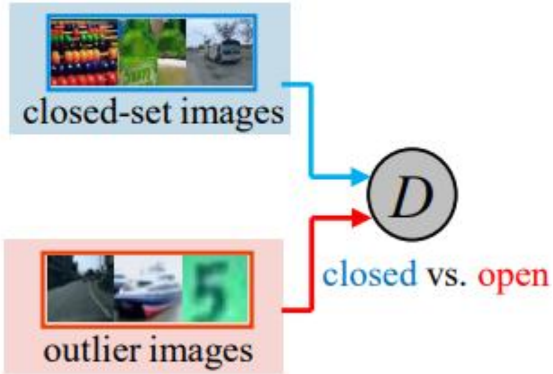


Approaches to open-set recognition

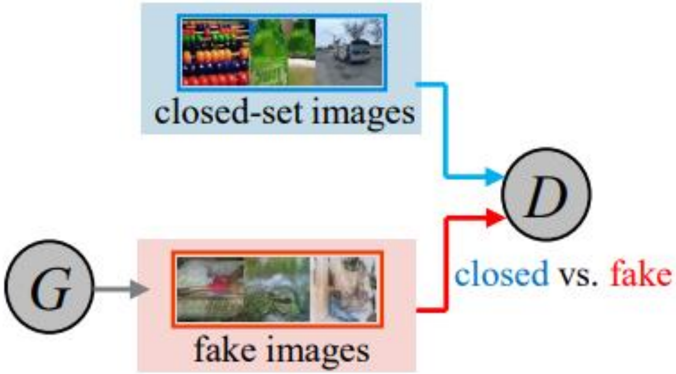
- Nearest Neighbor & Nearest Class/Cluster Mean (NCM)
- Max of softmax score & max of logit score
- Entropy
- Outlier exposure
- Outlier generation
- **Joint force with OpenGAN**

Open-set recognition with both synthetic data and real outlier data

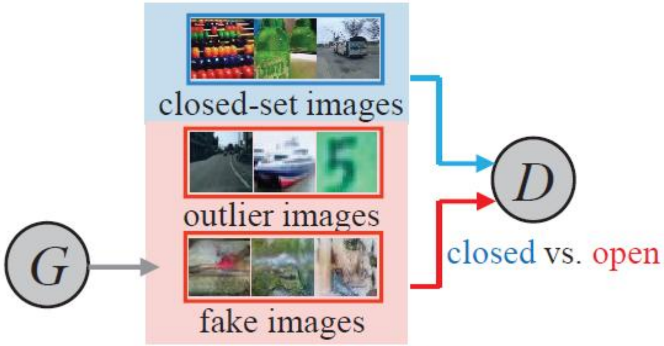
Use both outlier data and synthetic data as a better set of “pseudo” open-set training data



(b) Outlier Exposure



(a) GAN

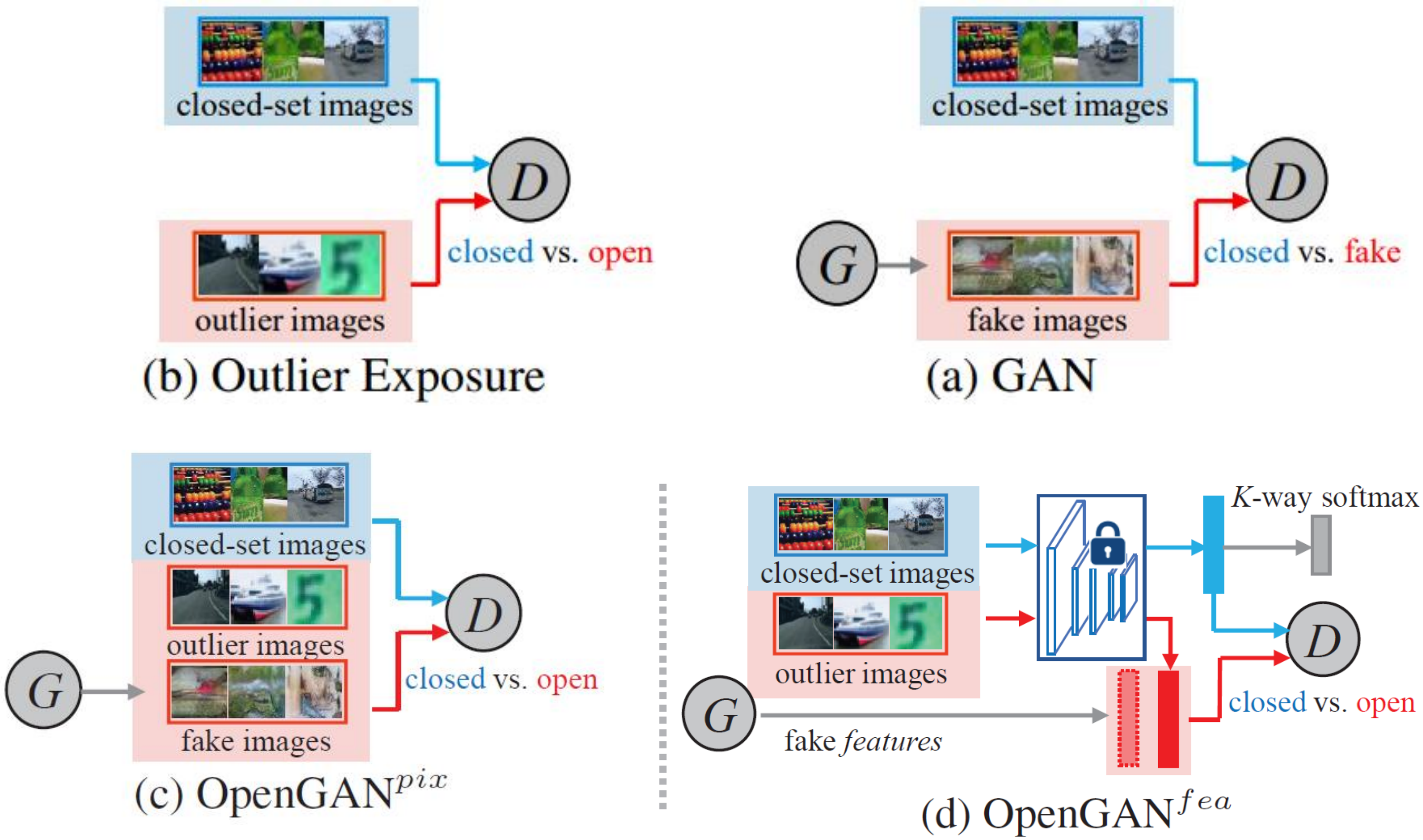


(c) OpenGAN^{pix}

Kong & Ramanan, "OpenGAN: Open-Set Recognition via Open Data Generation:, ICCV, 2021

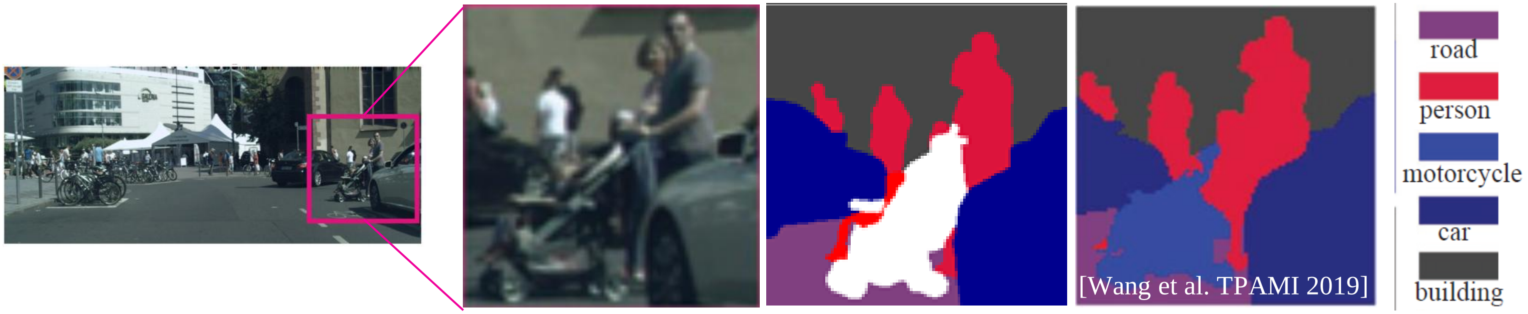
Open-set recognition with both synthetic data and real outlier data

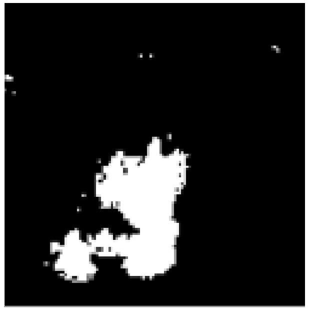
Use both outlier data and synthetic data as a better set of “pseudo” open-set training data



Kong & Ramanan, "OpenGAN: Open-Set Recognition via Open Data Generation:, ICCV, 2021

Open-set in autonomous driving





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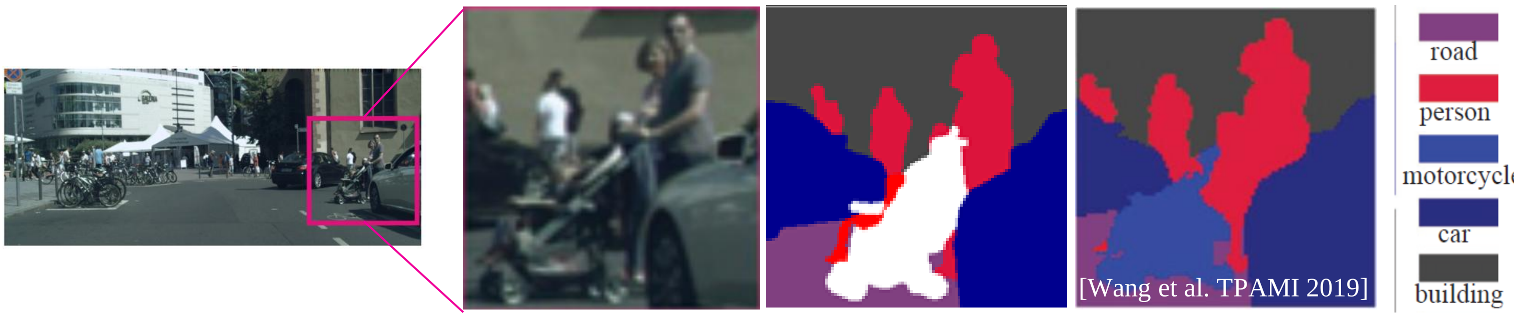
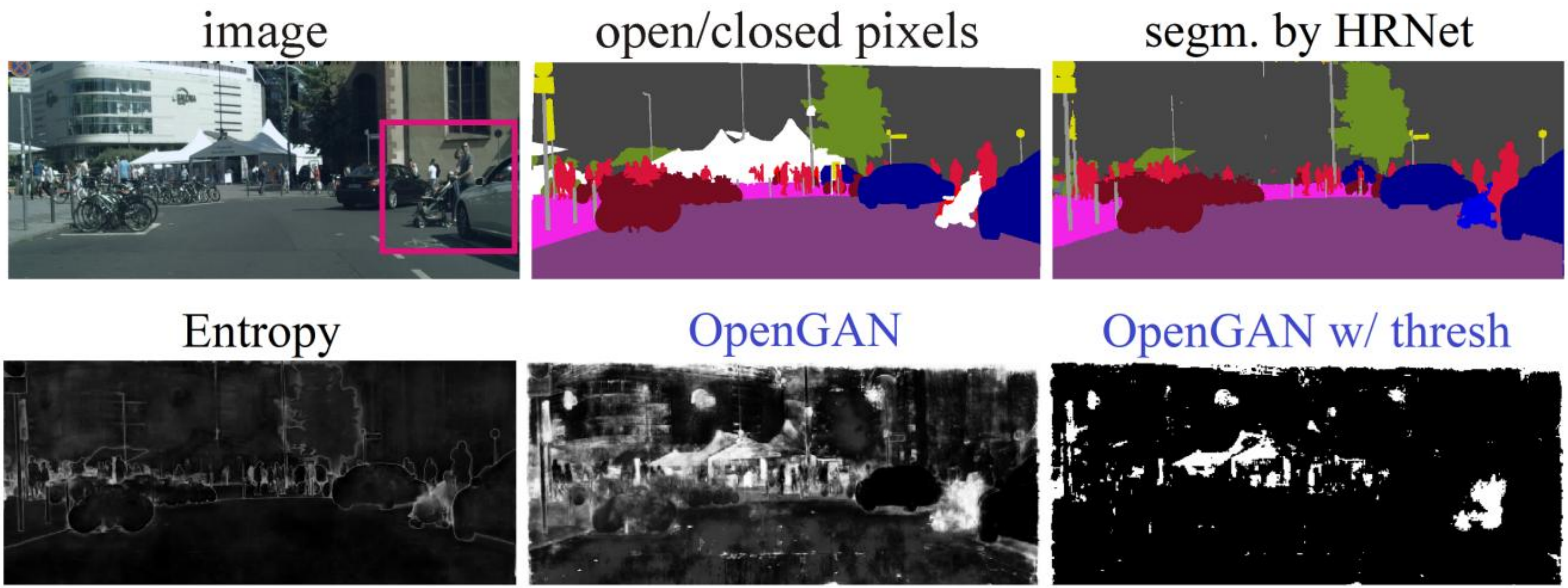
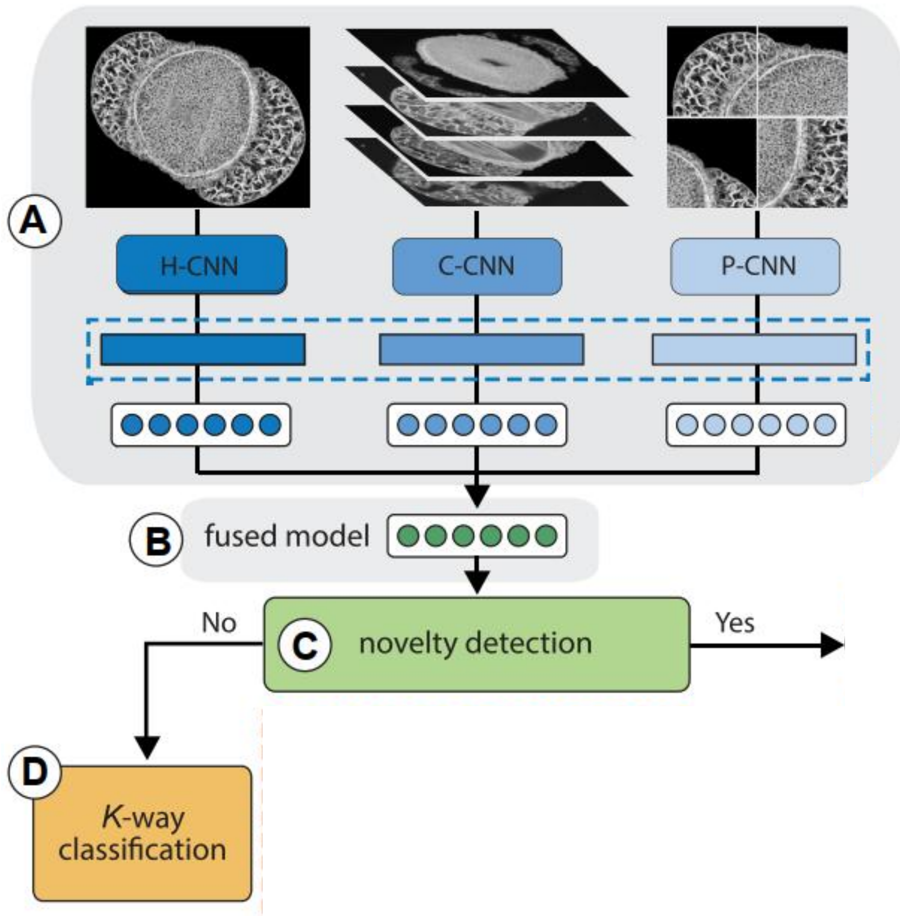




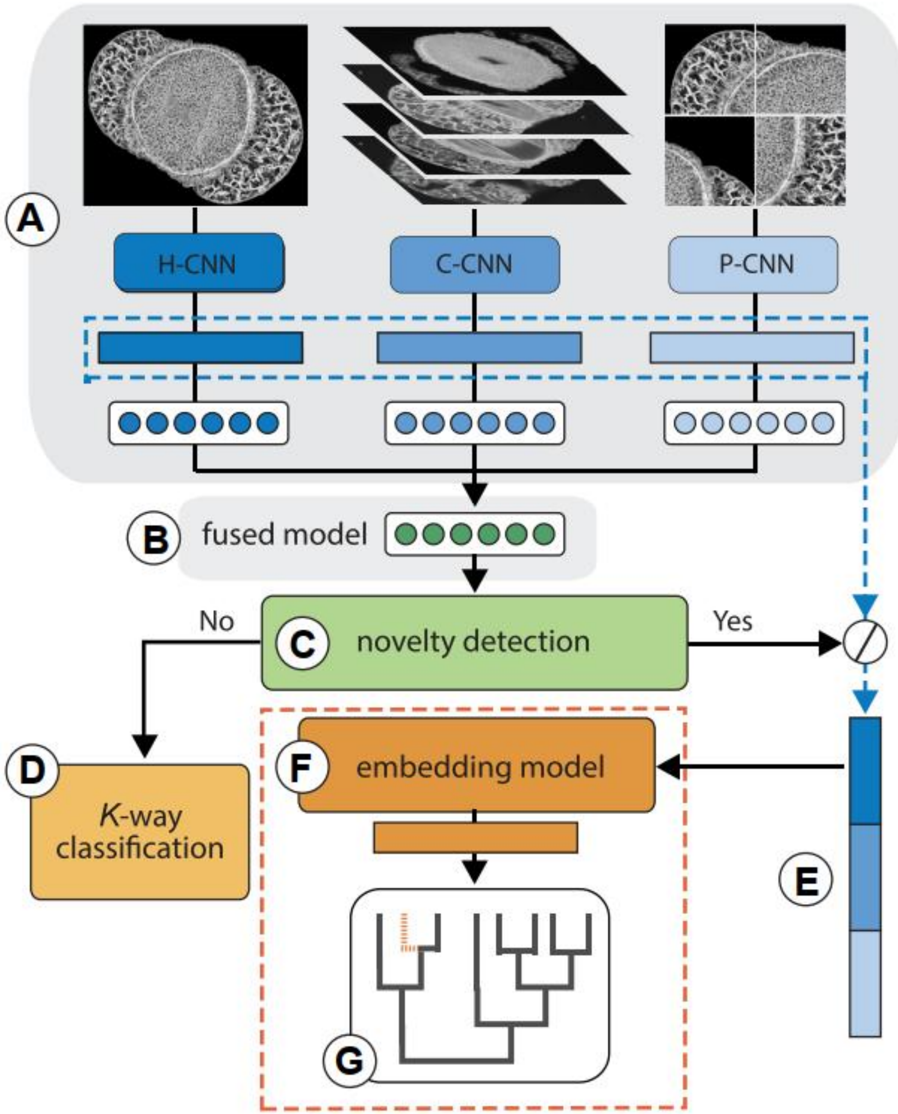
Open-set for novel species recognition



Adaïmé, et al., "Deep learning approaches to the phylogenetic placement of extinct pollen morphotypes", PNAS Nexus, 2024

Adaimé, et al., "Deep learning of fossil pollen morphology reveals 25,000 years of ecological change in East African grasslands", in submission

Open-set for novel species recognition



Adaïmé, et al., "Deep learning approaches to the phylogenetic placement of extinct pollen morphotypes", PNAS Nexus, 2024
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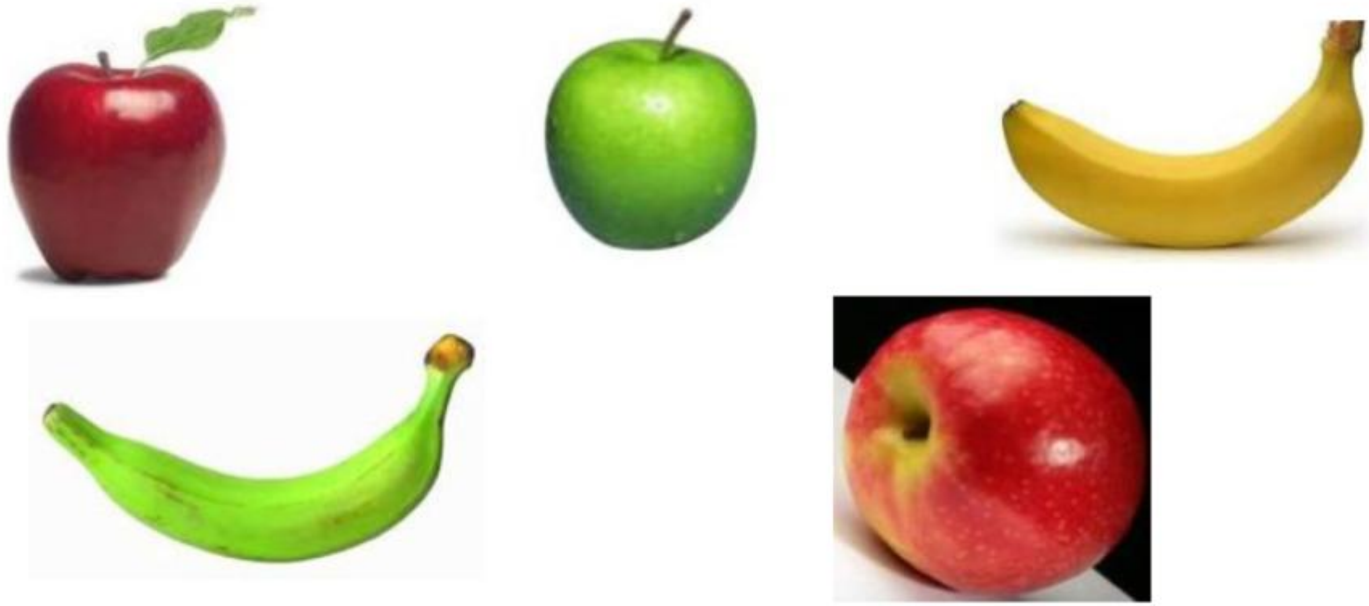
Outline

- Open-Set Recognition for Novel Species Detection
- **Clustering for Phylogenetic Reconstruction**

Approaches to clustering

- **Clustering**
- K-Means Clustering
- Agglomerative Clustering

Unsupervised Learning



Unsupervised learning: given data, i.e. examples, but no labels

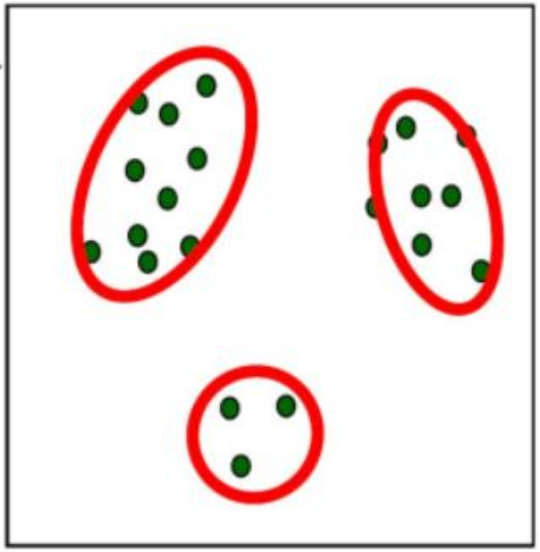
Unsupervised Learning

Supervised learning

- Predict target value ("y") given features ("x")

Unsupervised learning

- Understand patterns of data (just "x")
- Useful for many reasons
  - **Data mining ("explain")**
  - **Missing data values ("impute")**
  - **Representation (feature generation or selection)**



One example: clustering

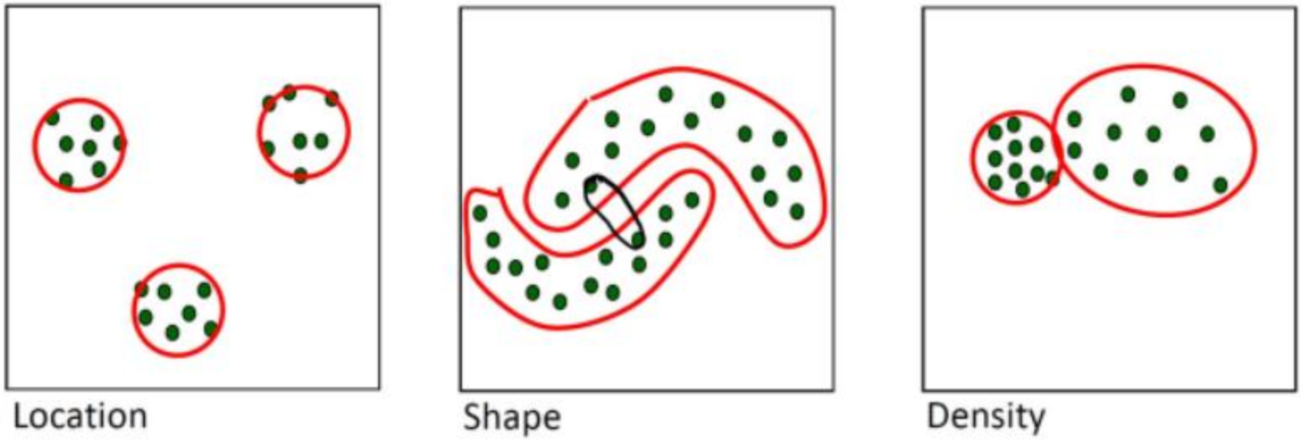
- Describe data by discrete "groups" with some characteristics

Clustering

Clustering describes data by “groups”

The meaning of “groups” may vary by data!

Examples



Approaches to clustering

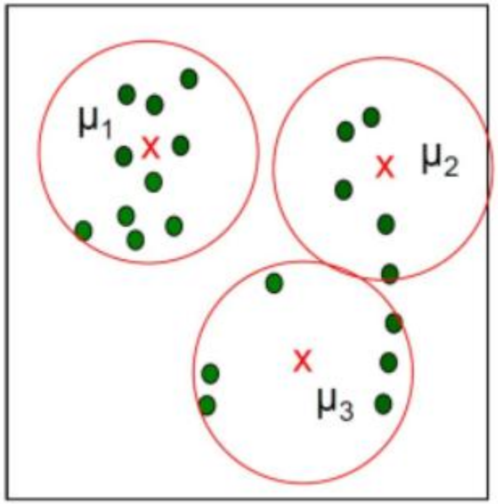
- Clustering
- **K-Means Clustering**
- Agglomerative Clustering

K-Means Clustering

A simple clustering algorithm

Iterate between

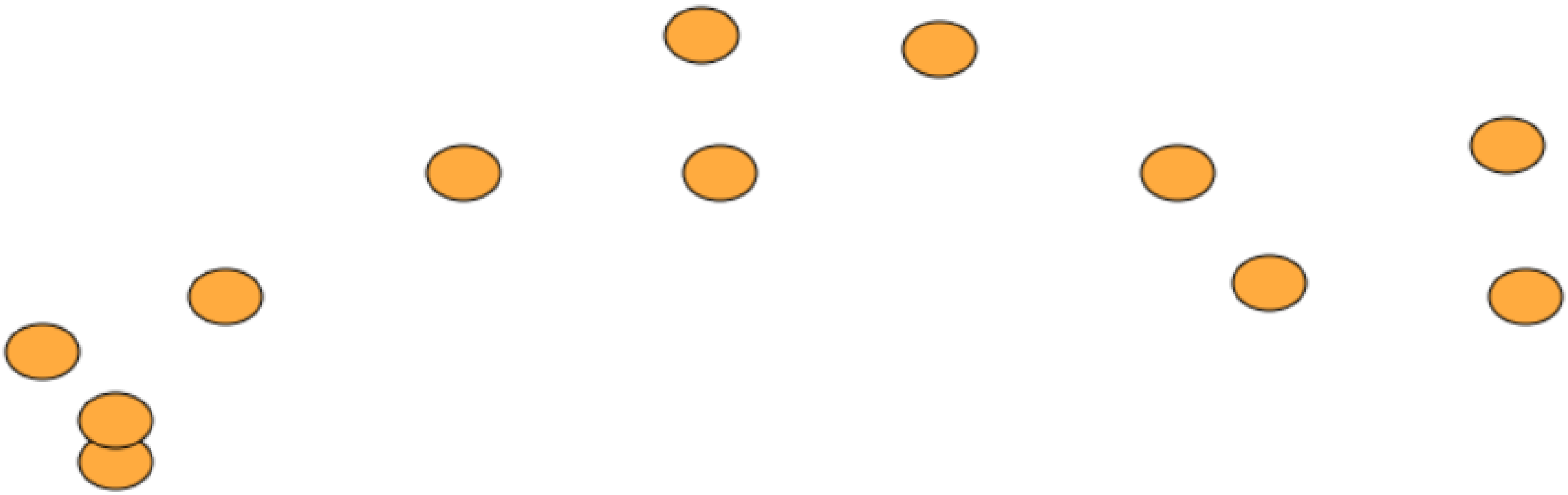
- Updating the assignment of data to clusters
- Updating the cluster's summarization



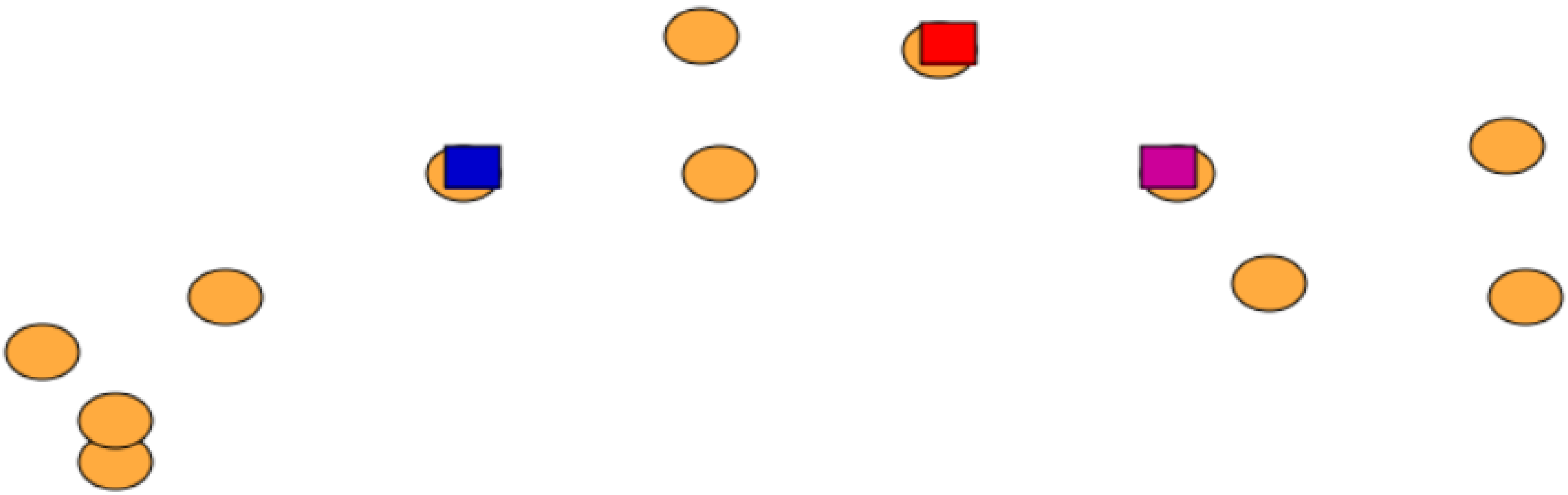
Notation:

1. Data example i has features x_i
2. Assume K clusters, (K=3)
3. Each cluster c "described" by a center μ_c
4. Each cluster will "claim" a set of nearby points

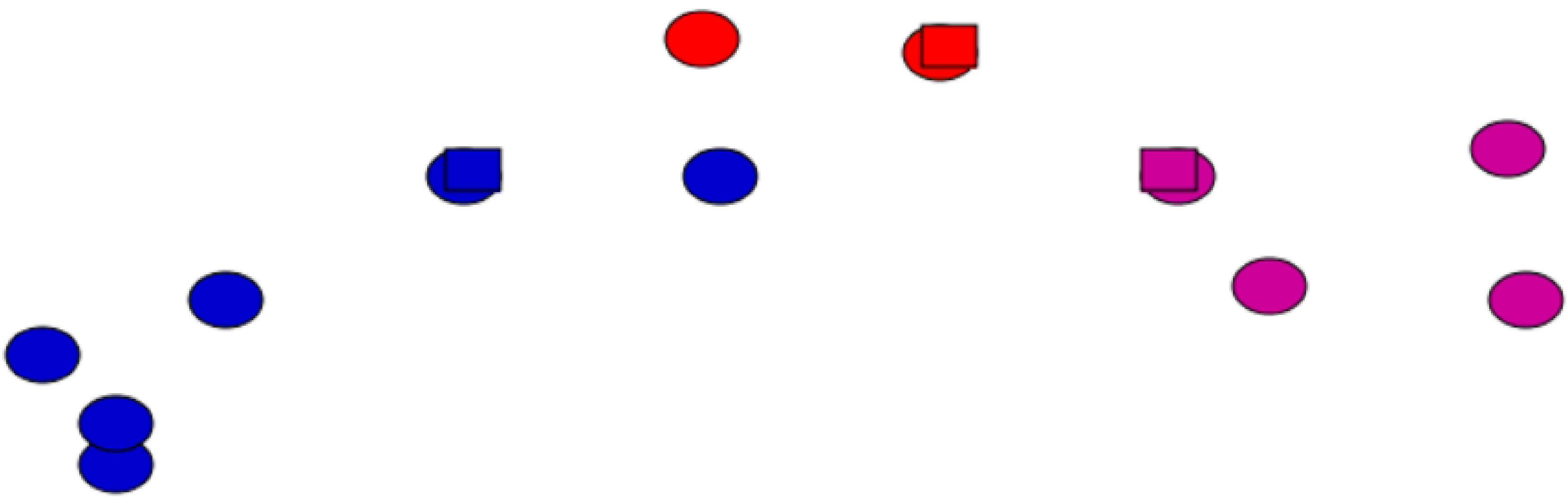
K-Means: an example



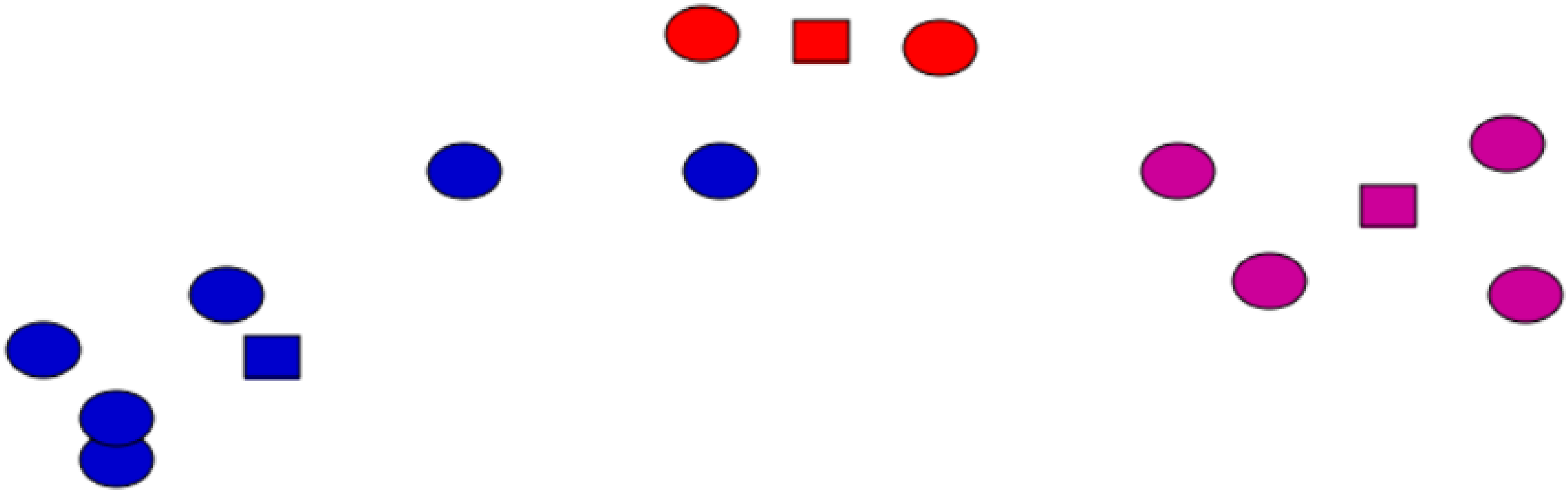
K-Means: Initialize Centers Randomly



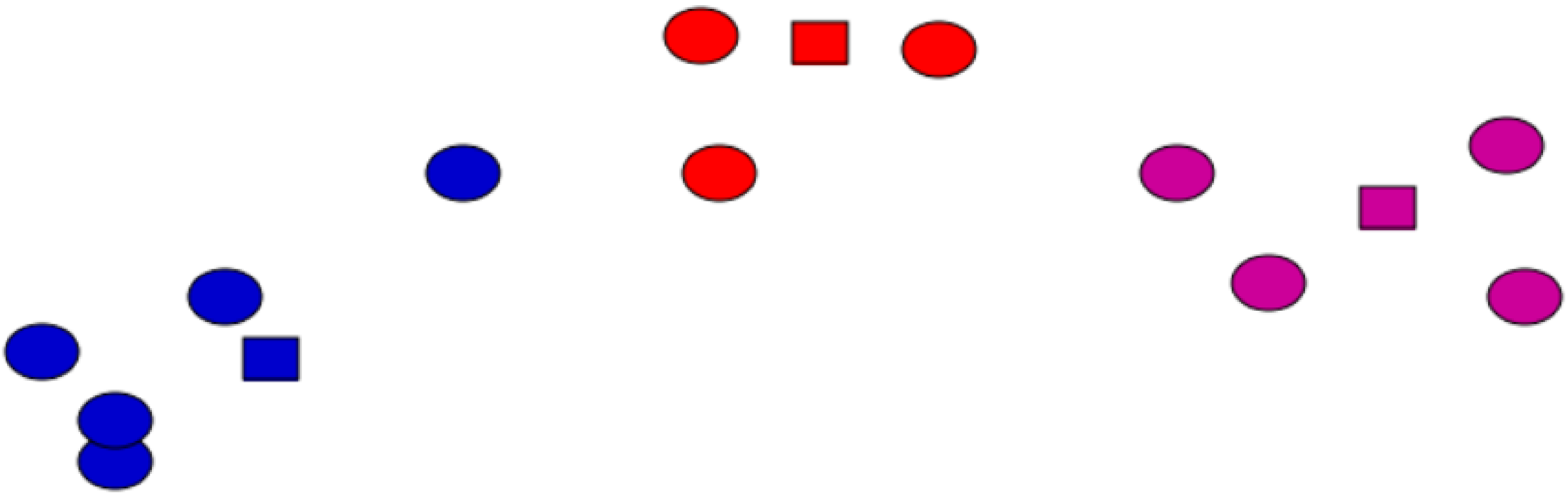
K-Means: Assign Points to Nearest Center



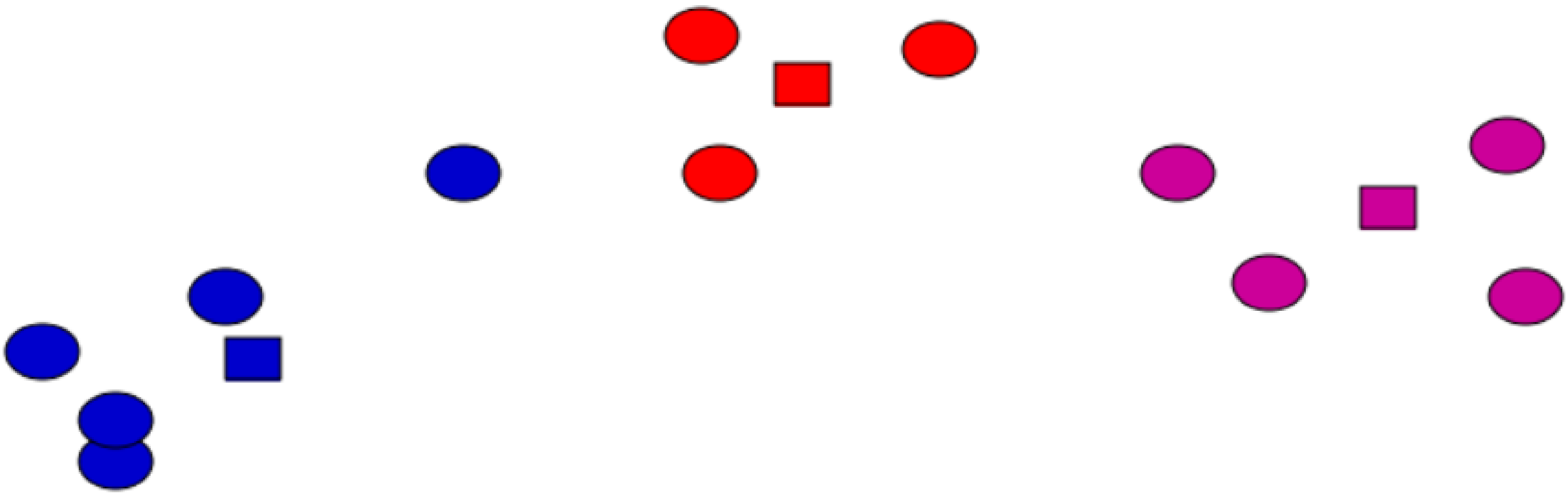
K-Means: Re-adjust centers



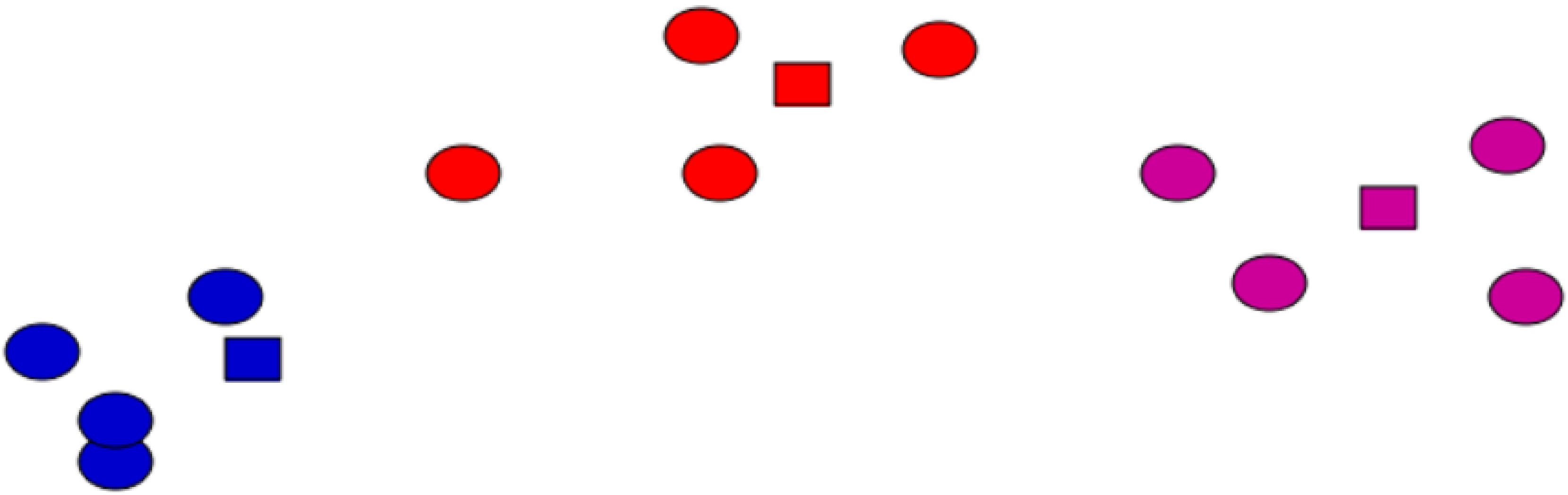
K-Means: Assign Points to Nearest Center



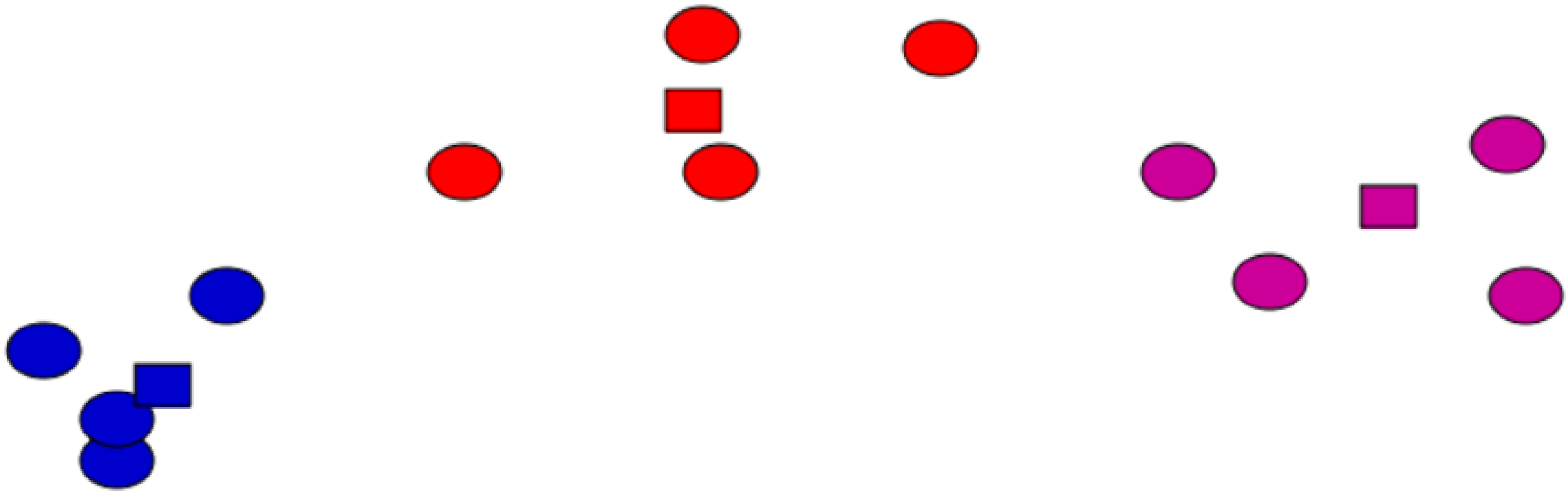
K-Means: Re-adjust centers



K-Means: Assign Points to Nearest Center



K-Means: Re-adjust centers



K-Means: Assign Points to Nearest Center

K-Means Clustering

Optimizing the cost function: $$C(\underline{z}, \underline{\mu}) = \sum_i \|x_i - \mu_{z_i}\|^2$$

Iterate until convergence:

(A) For each datum, find the closest cluster: (z_i denotes cluster membership)

$$z_i = \arg\min_c \|x_i - \mu_c\|^2 \qquad \forall i$$

(B) Set each cluster to the mean of all assigned, for each cluster c:

$$\mu_c = \frac{1}{m_c} \sum_{i \in S_c} x_i \qquad\qquad S_c = \{i : z_i = c\},\ m_c = |S_c|$$

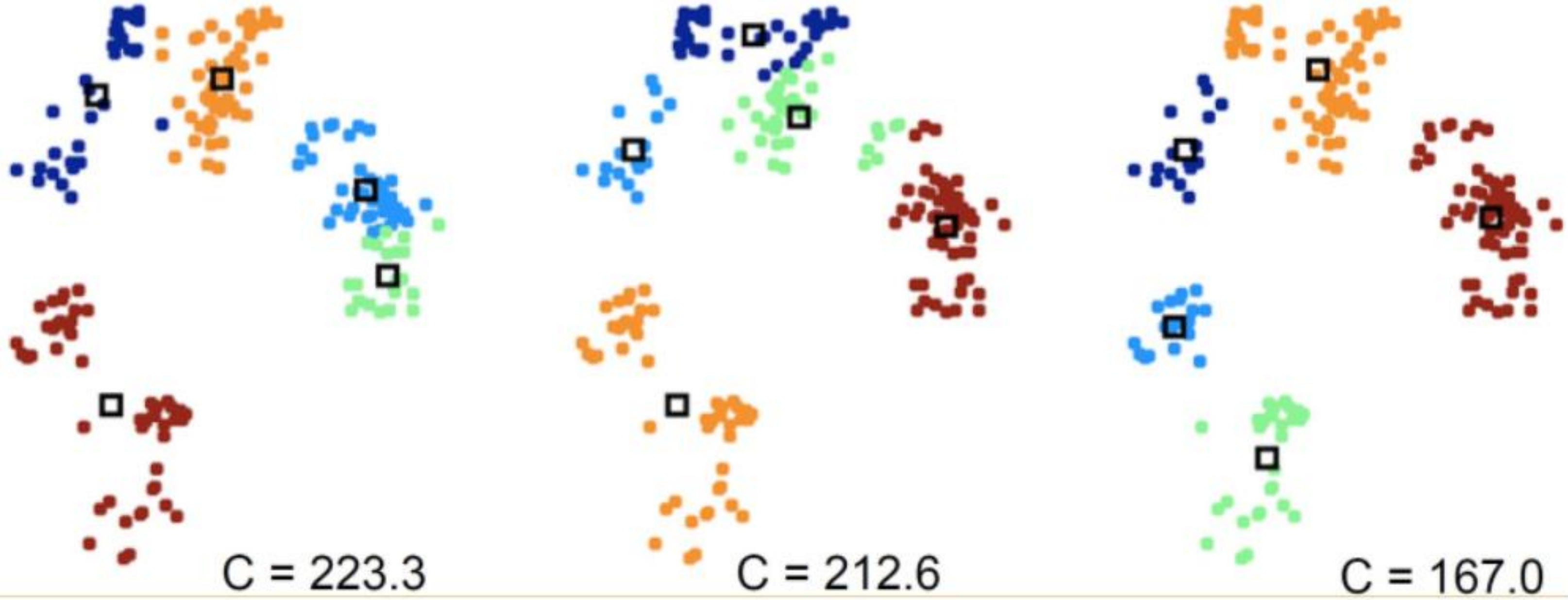
Initialization

Optimizing the cost function: $$C(\underline{z}, \underline{\mu}) = \sum_i \|x_i - \mu_{z_i}\|^2$$

Multiple local optima, depending on initialization

Try different (randomized) initializations

Can use cost C to decide which we prefer

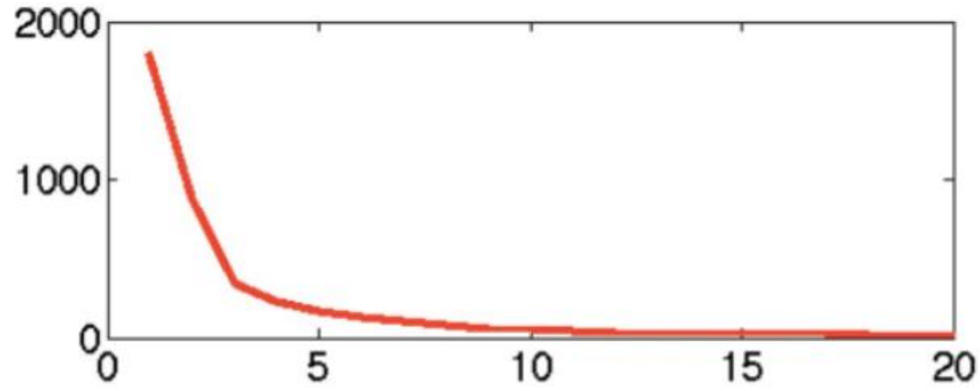


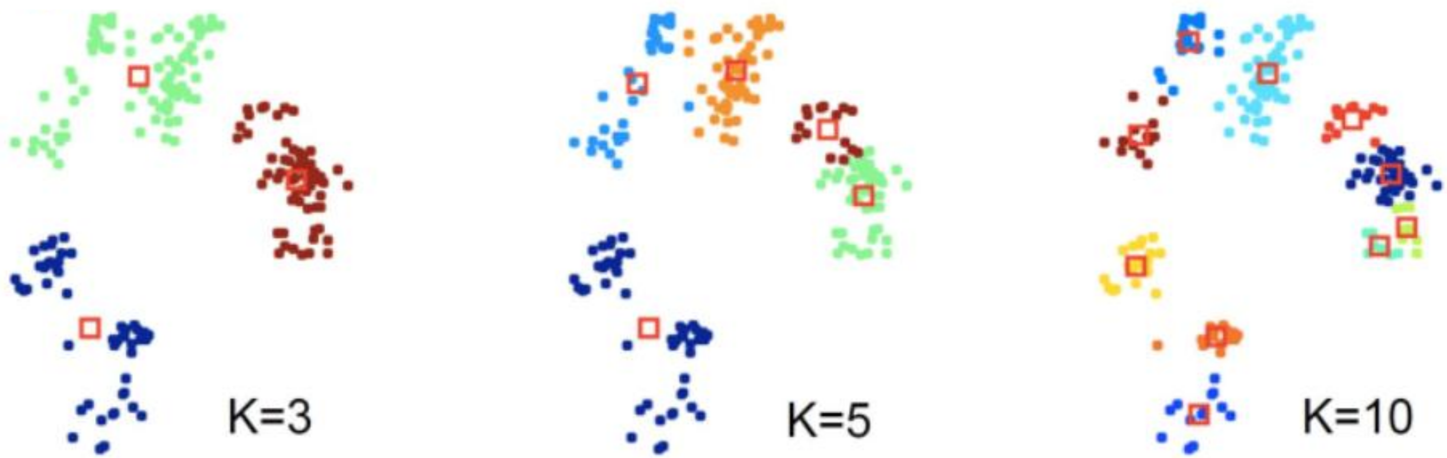
Choosing Number of Clusters

With cost function

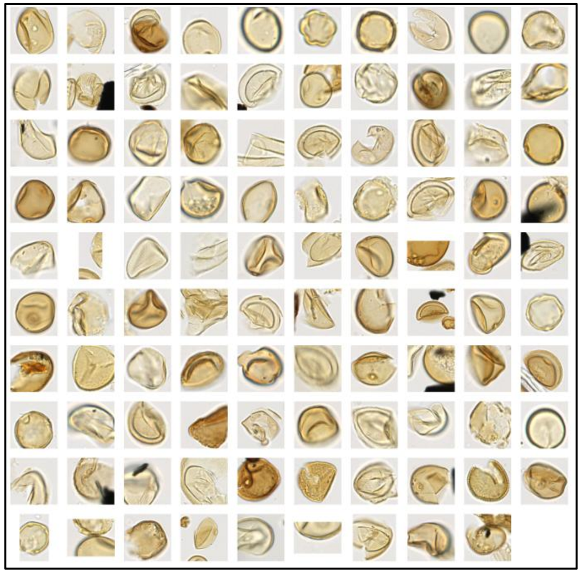
$$C(\underline{z}, \underline{\mu}) = \sum_i \|x_i - \mu_{z_i}\|^2$$

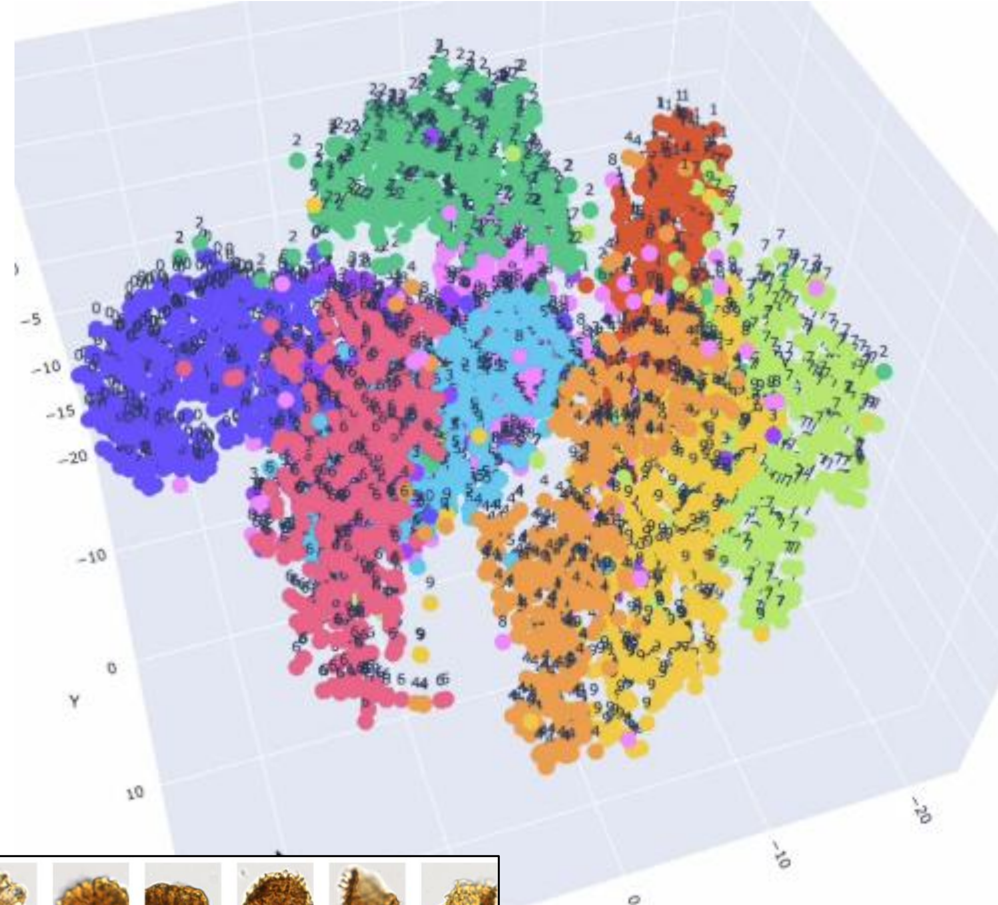
What is the optimal value K?
Cost always decreases with K!
A model complexity issue…

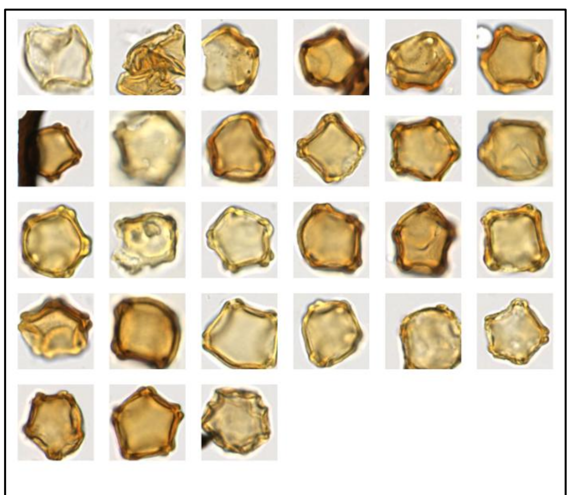




Clustering to Facilitate Annotation







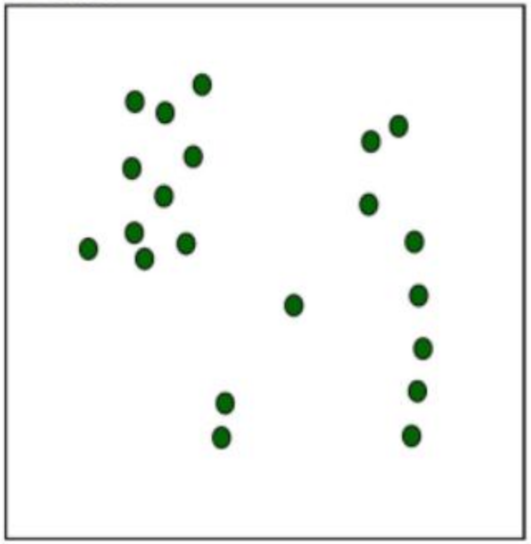
Approaches to clustering

- Clustering
- K-Means Clustering
- **Agglomerative Clustering**

Hierarchical Agglomerative Clustering

Initially, every datum is a cluster

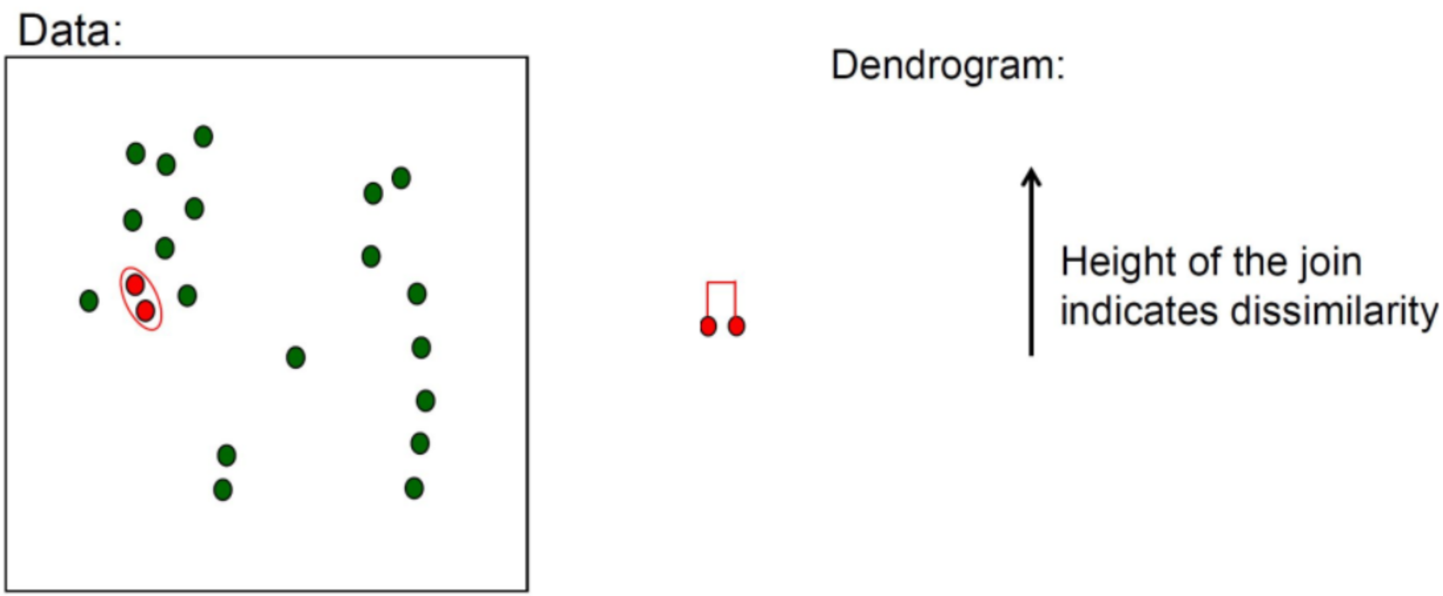
Data:



- A simple clustering algorithm
- Define a distance (or dissimilarity) between clusters (we'll return to this)
- Initialize: every example is a cluster
- Iterate:
  - Compute distances between all clusters (store for efficiency)
  - Merge two closest clusters
- Save both clustering and sequence of cluster operations
- Dendrogram

Iteration 1

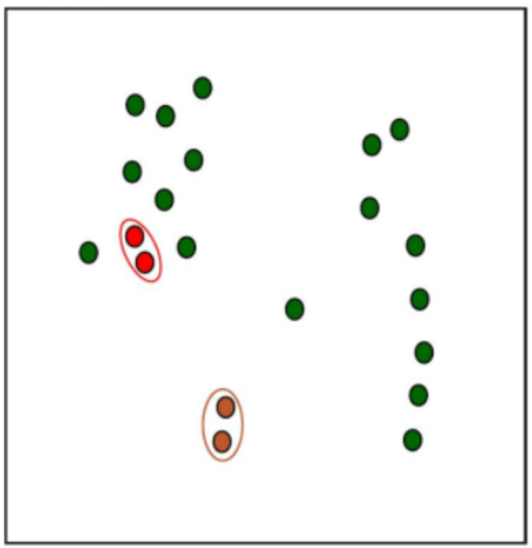
Builds up a sequence of clusters (“hierarchical”)



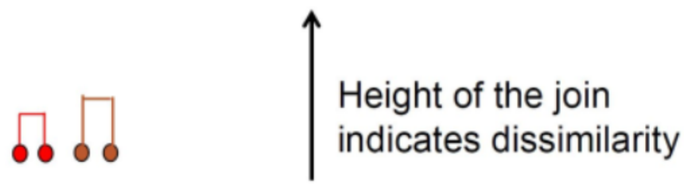
Iteration 2

Builds up a sequence of clusters ("hierarchical")

Data:

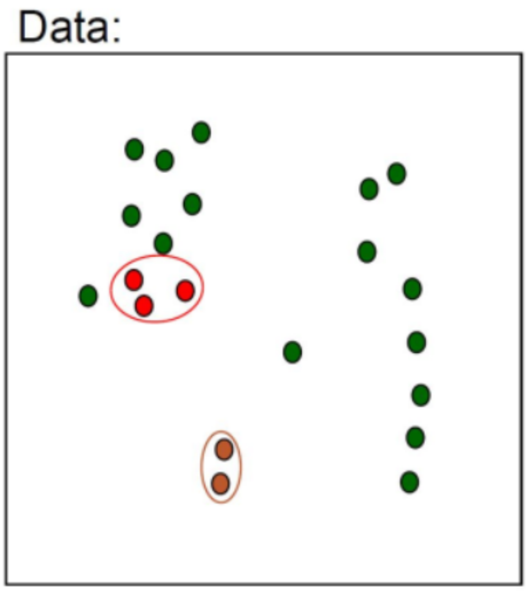


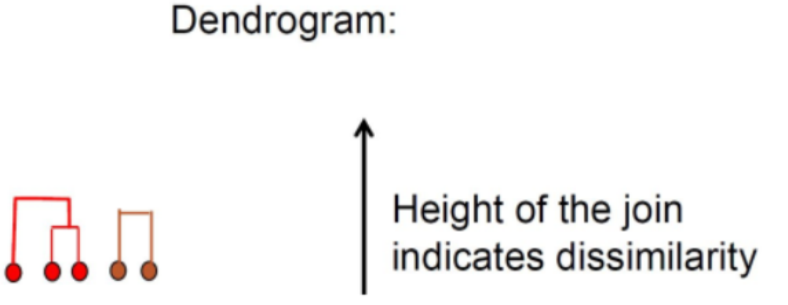
Dendrogram:



Iteration 3

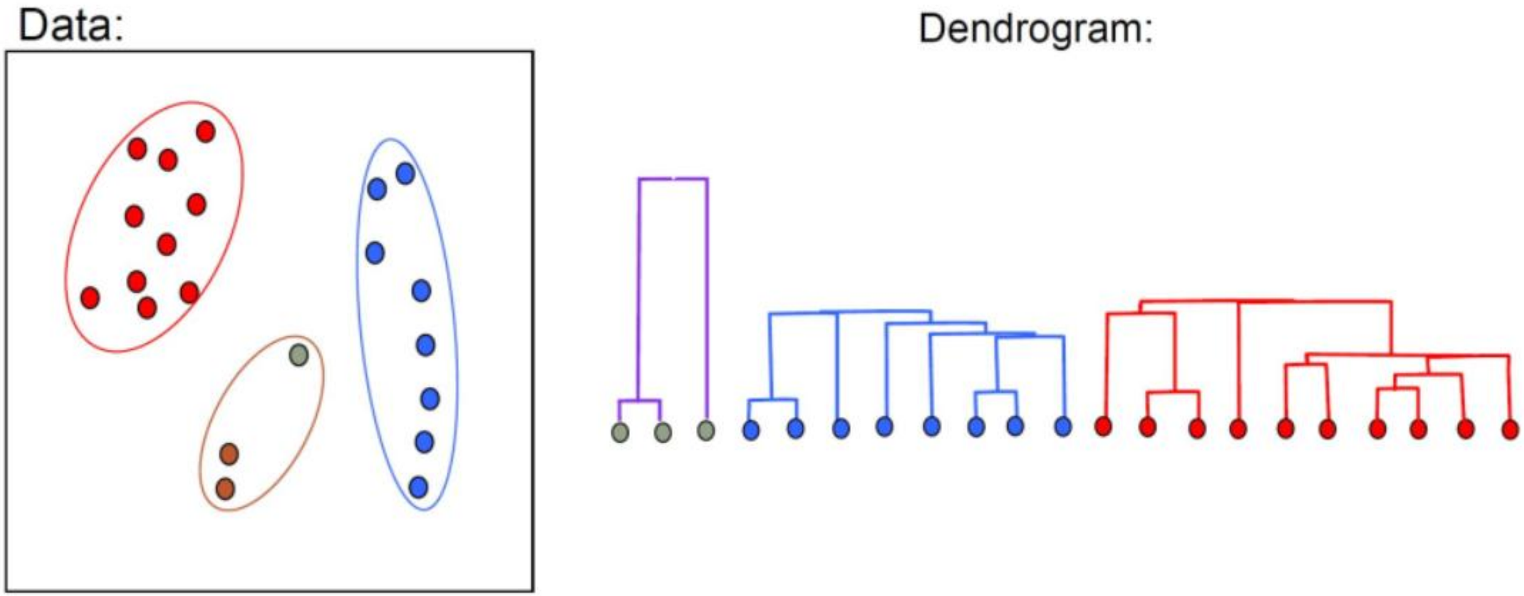
Builds up a sequence of clusters (“hierarchical”)





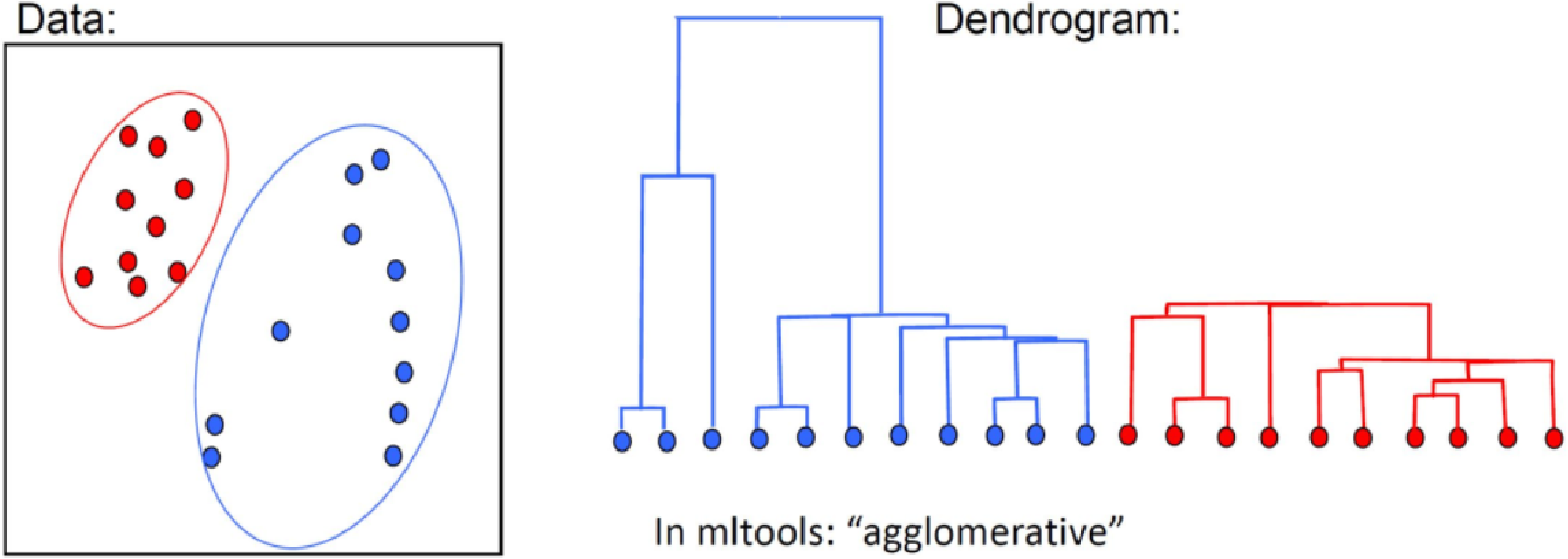
Iteration m-3

Builds up a sequence of clusters ("hierarchical")



Iteration m-2

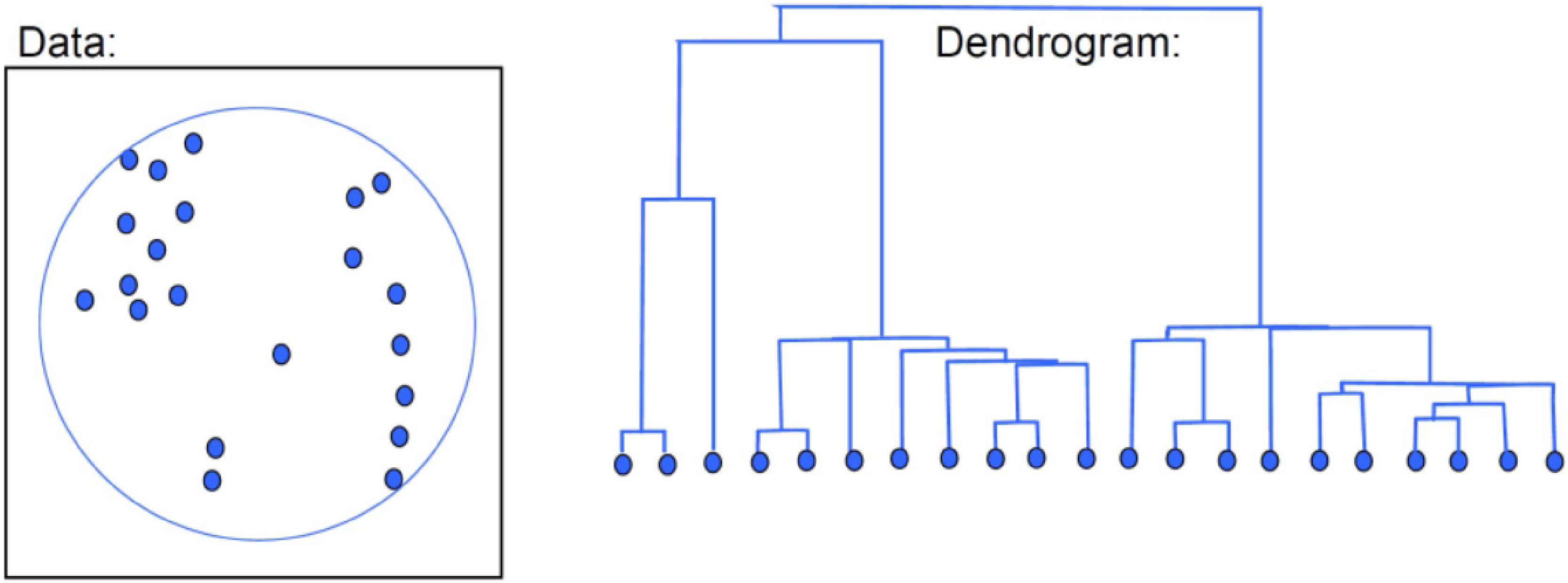
Builds up a sequence of clusters ("hierarchical")



In mltools: "agglomerative"

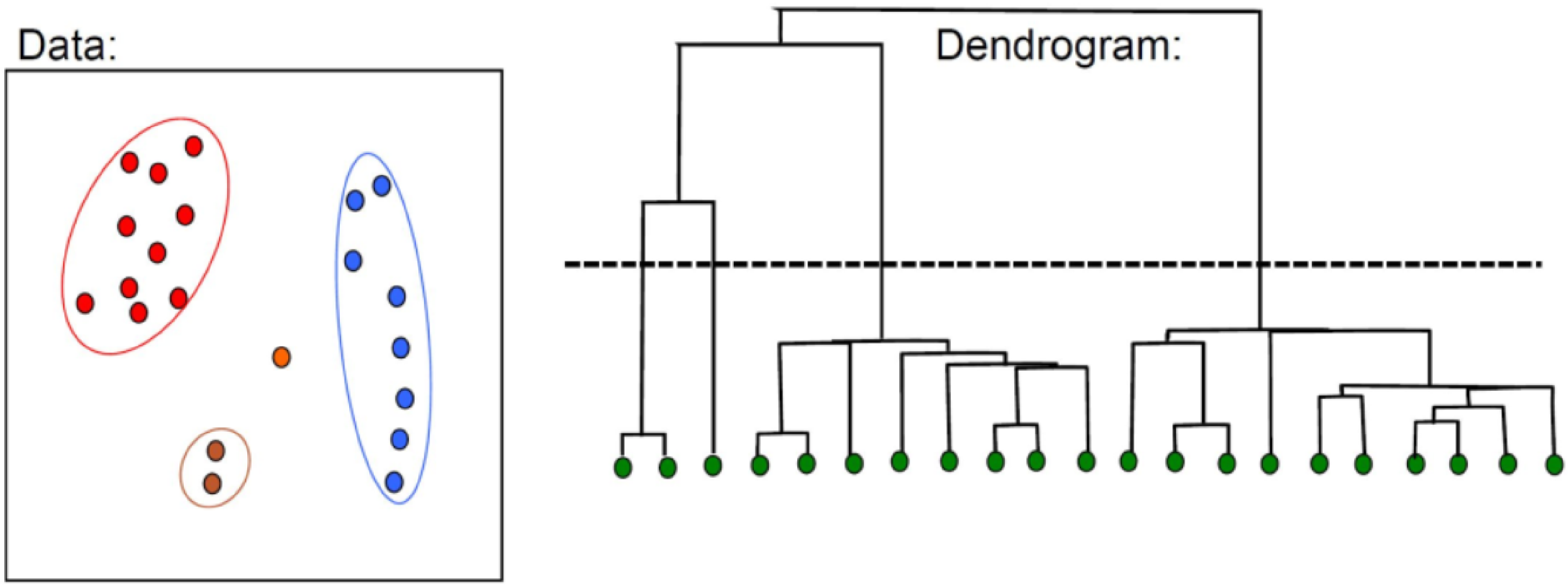
Iteration m-1

Builds up a sequence of clusters ("hierarchical")

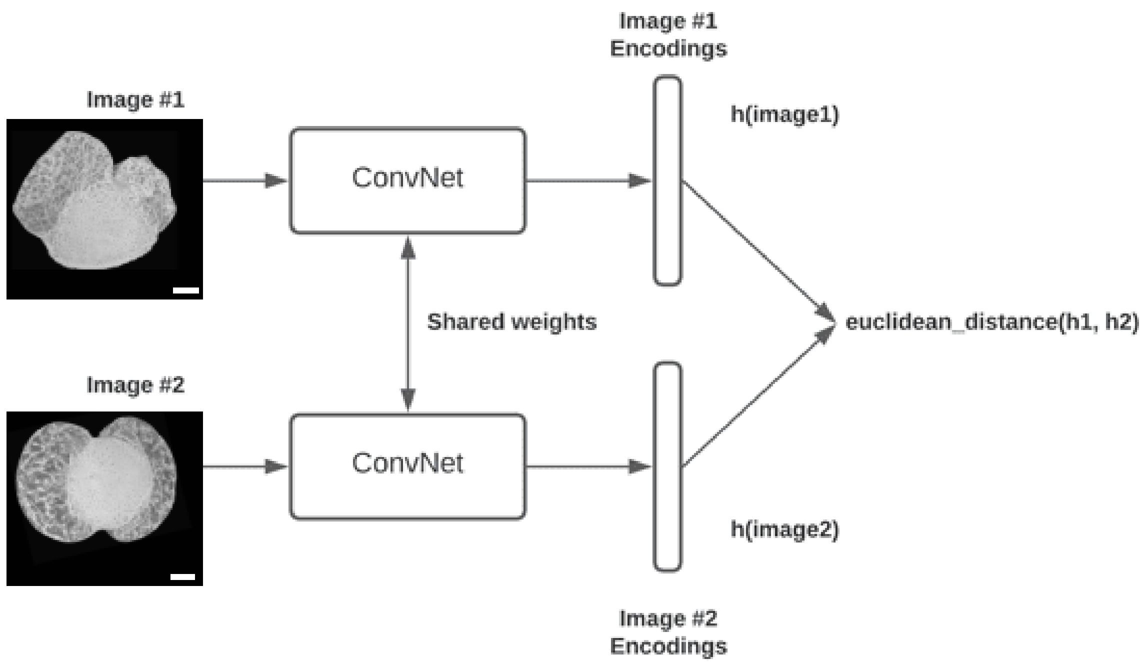


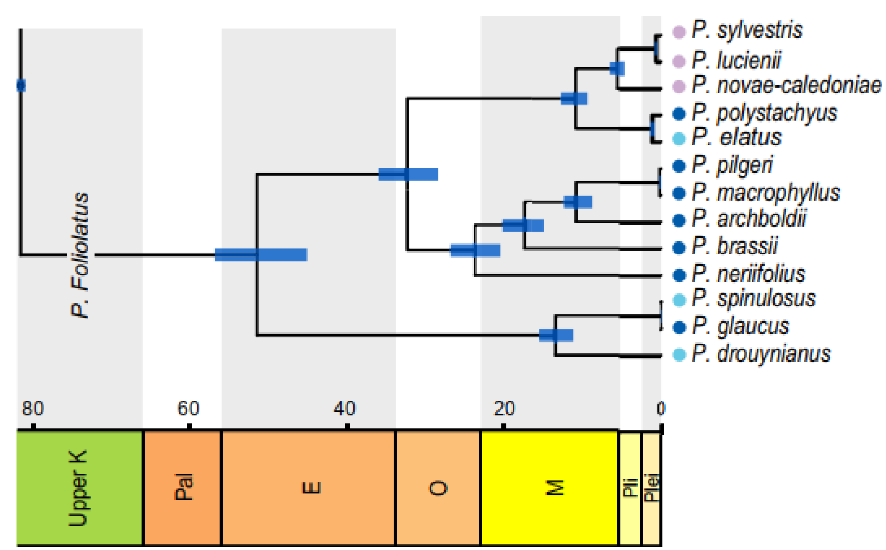
From Dentrogram to Clusters

Given the sequence, can select a number of clusters or a dissimilarity threshold:



Clustering for Phylogenetic Reconstruction





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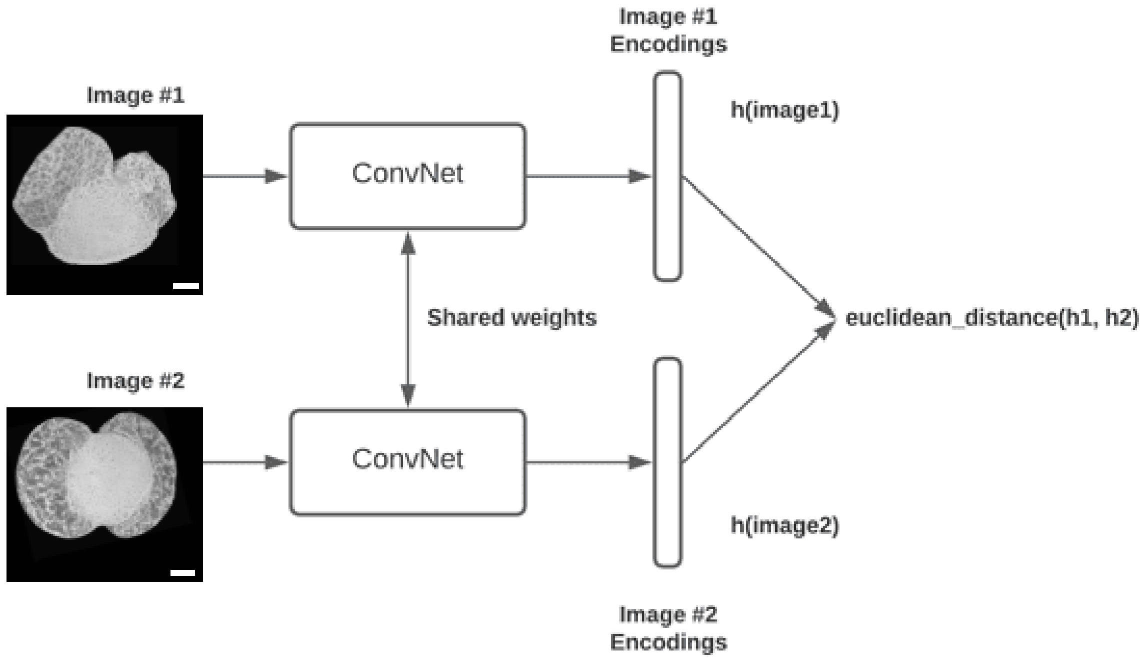

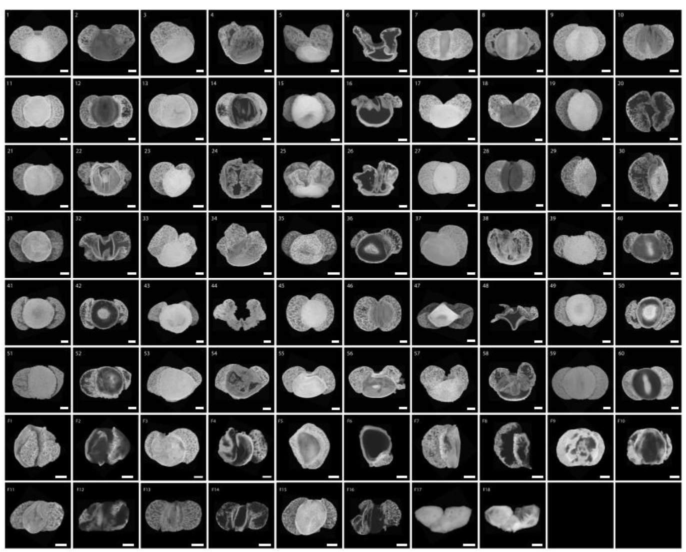

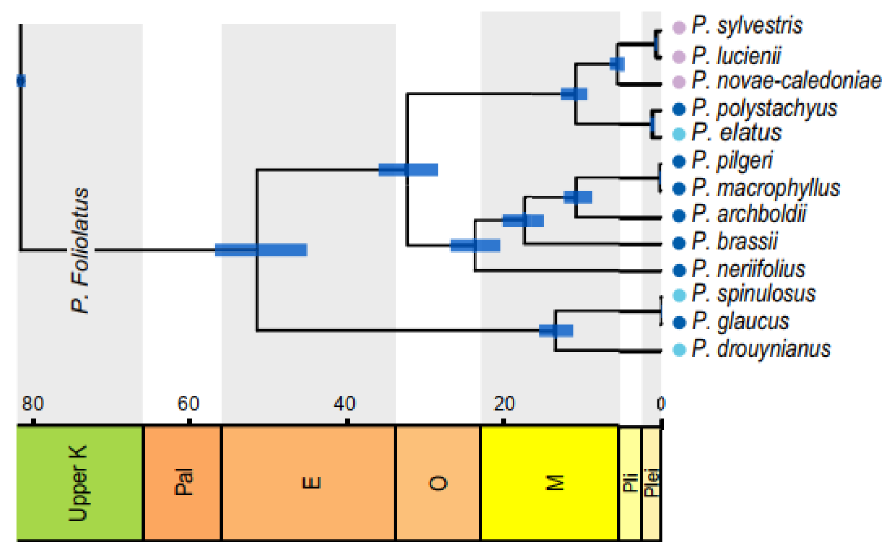

Clustering for Phylogenetic Reconstruction

- Initialize: every example is a cluster
- Iterate:
    - Compute distances between all clusters (store for efficiency)
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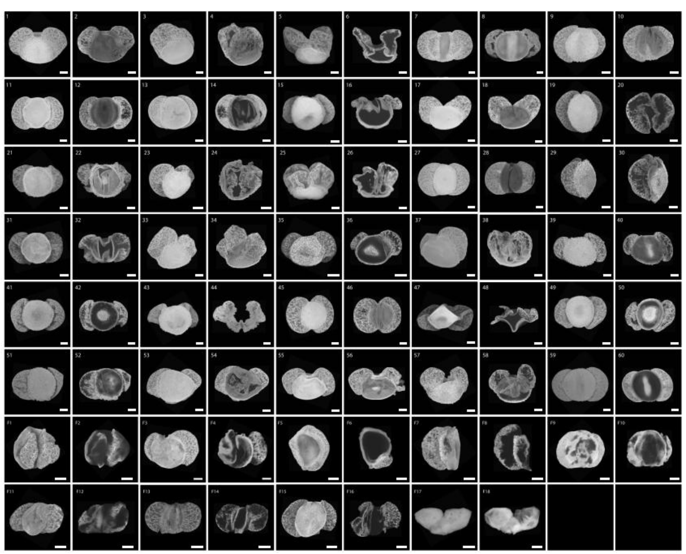
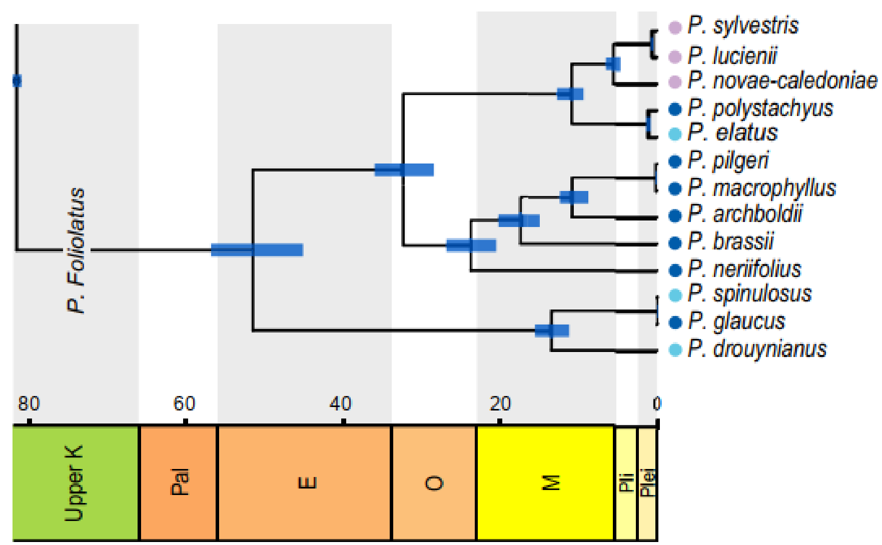
- Save both clustering and sequence of cluster operations
- Dendrogram





Clustering for Phylogenetic Reconstruction

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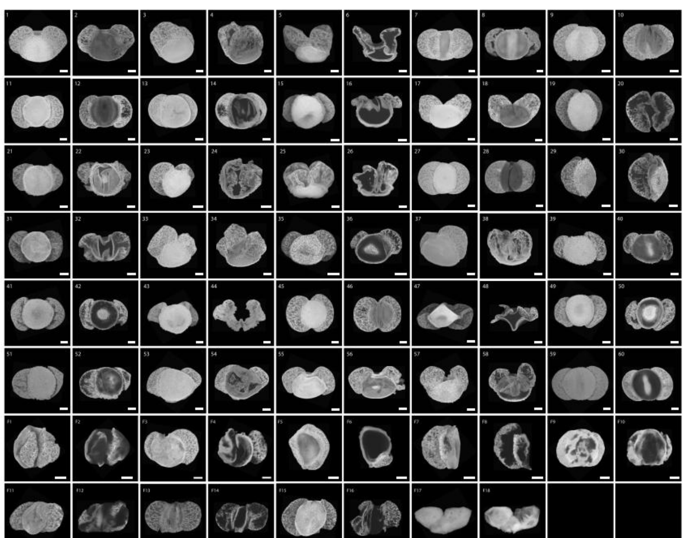
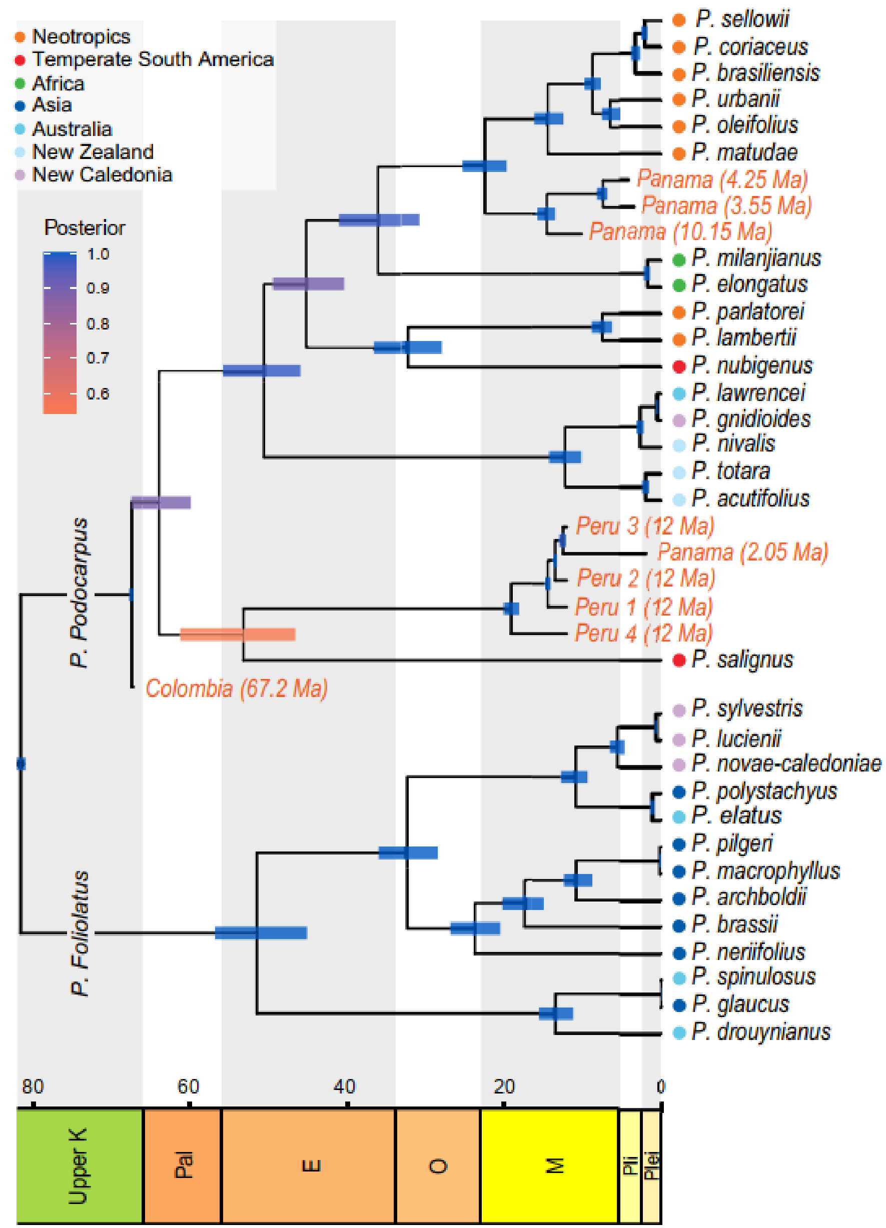




Approaches to clustering

- Clustering
- K-Means Clustering
- Agglomerative Clustering

Outline

- Open-Set Recognition for Novel Species Detection
- Clustering for Phylogenetic Reconstruction

Roadmap

Toolboxes:

- platform: Python, Jupyter Notebook
- basics: sklearn, skimage, numpy, scipy, matplotlib
- for deep learning: PyTorch
- Teaching materials are available at https://github.com/aimerykong/MPC2026-tutorial

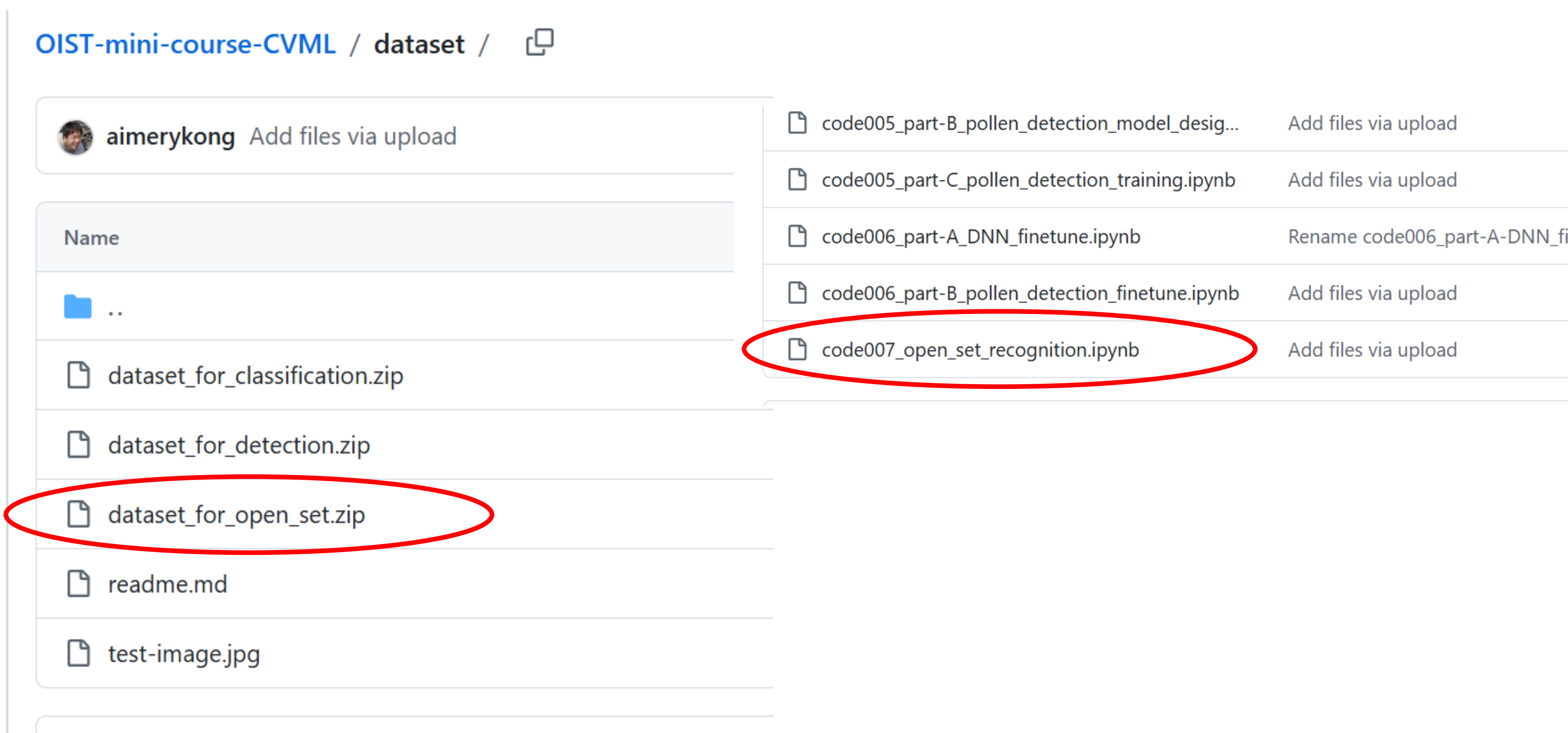
Lecture	Topics
1. Introduction	machine learning, computer vision, deep learning, artificial intelligence
2. Basics via Classification	Nearest Neighbor, Perceptron, Logistic Regression, cross entropy, feature, gradient descent, etc.
3. Deep Neural Networks	Multi-Layer Perceptron, feature learning, backpropagation, CNN, Transformer
4. Hands-On Practice: Basics	pollen data preparation, visualization, nearest neighbor, linear classifier, CNN
5. Segmentation and Detection	semantic segmentation, instance segmentation, U-Net, pixel grouping, mean-shift
6. Foundation Models	CLIP, Segment Anything Model, DINOv2, GroundingDINO, zero-shot learning
7. Advanced Topics	open-set recognition, novel species recognition, phylogentic reconstruction via clustering
8. Hands-On Practice: Advanced Topics	finetuning, pollen detection/segmentation

Hands-on exercise

https://github.com/aimerykong/OIST-mini-course-CVML

Today’s coding exercise:

- Train a deep neural network for pollen classification
- Use the trained model to detect open-set examples in test time
- Download the zip file “dataset_for_open_set.zip” and upload it to your gdrive
- Start with the jupyter notebook “code007_open_set_recognition.ipynb” in colab



Can we avoid specifying the number K in GMM and K-means?

Mean shift

Article Talk

From Wikipedia, the free encyclopedia

Mean shift is a non-parametric feature-space mathematical analysis technique for locating the maxima of a density function, a so-called mode-seeking algorithm.[1] Application domains include cluster analysis in computer vision and image processing.[2]

Mean-Shift clustering

Mean shift

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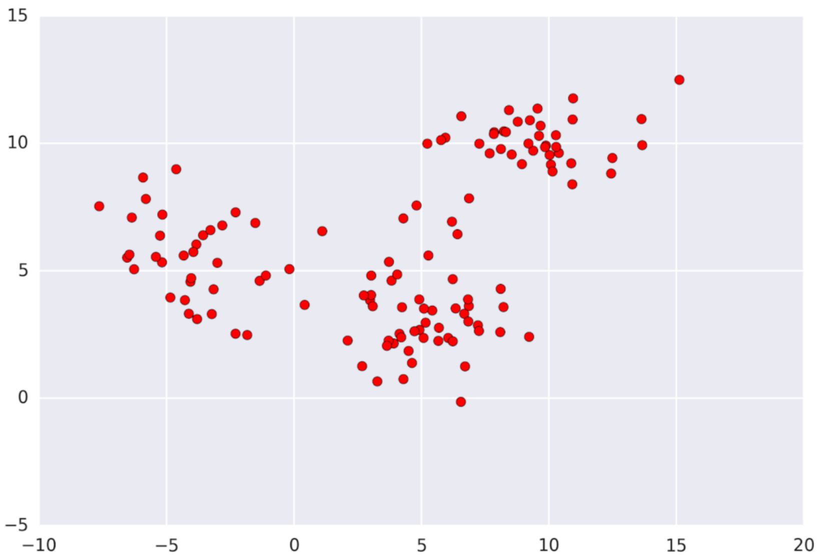


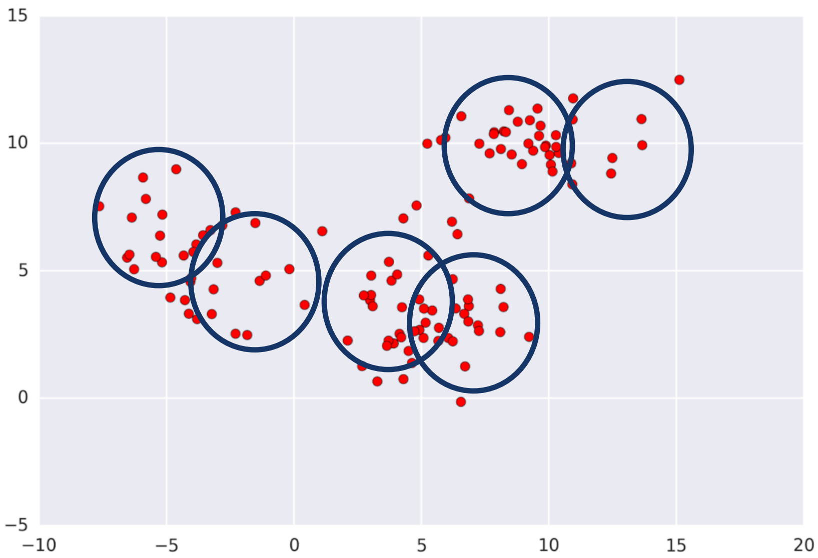
Mean-Shift clustering

Mean shift

Article Talk

From Wikipedia, the free encyclopedia

Mean shift is a non-parametric feature-space mathematical analysis technique for locating the maxima of a density function, a so-called mode-seeking algorithm.[1] Application domains include cluster analysis in computer vision and image processing.[2]



Mean-Shift clustering

Mean shift

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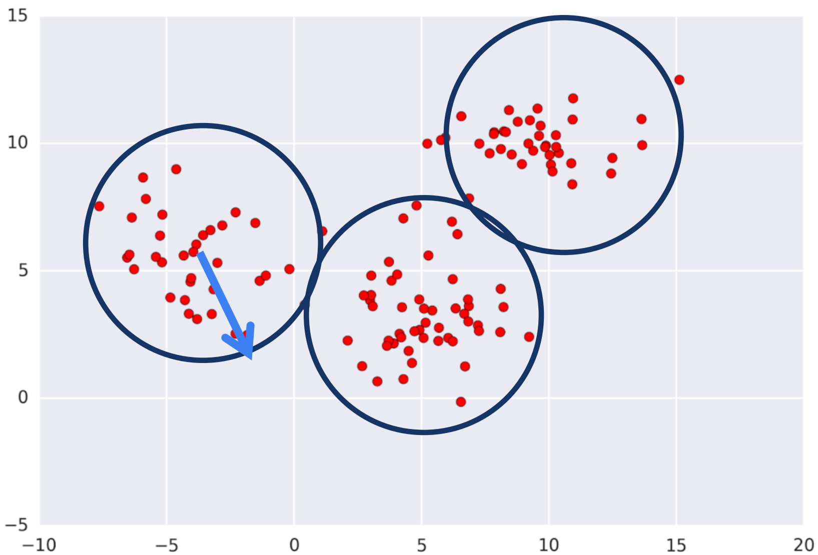

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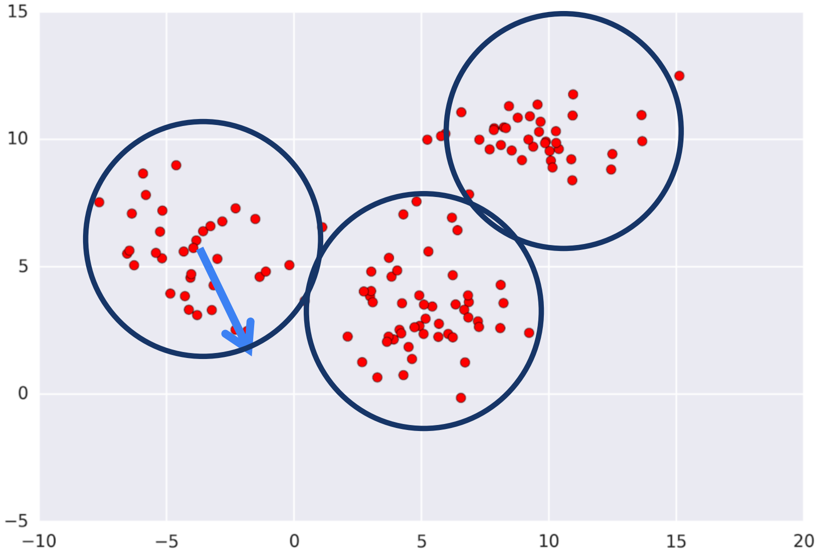

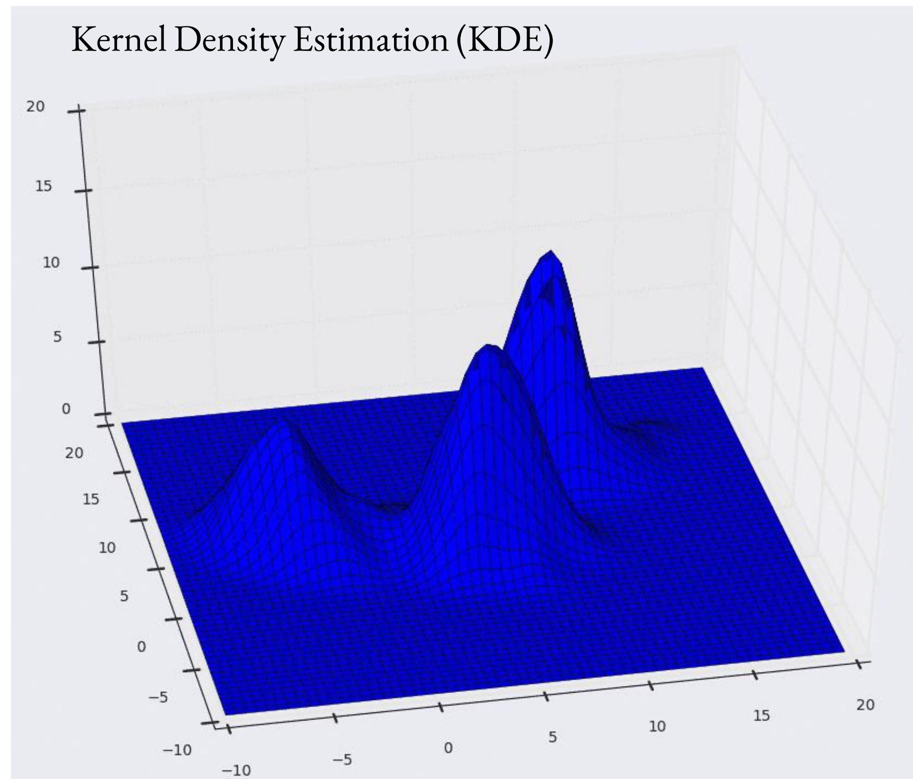

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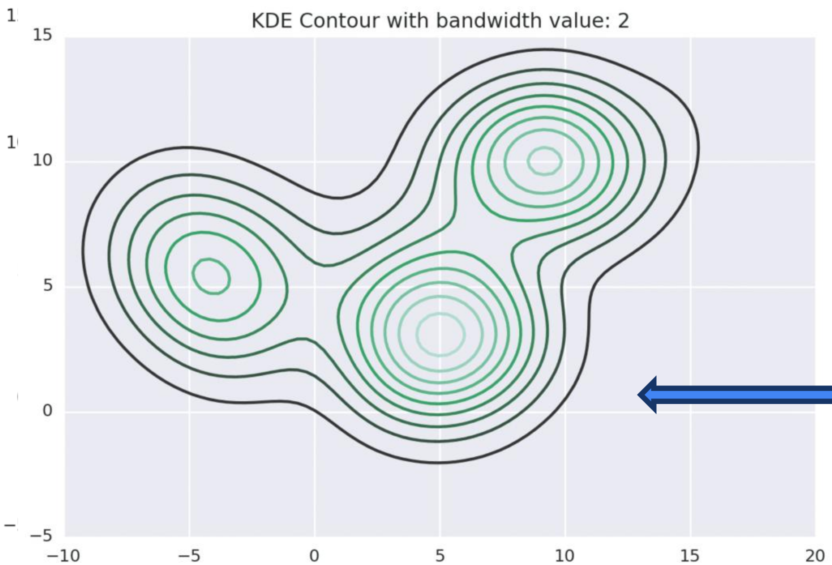

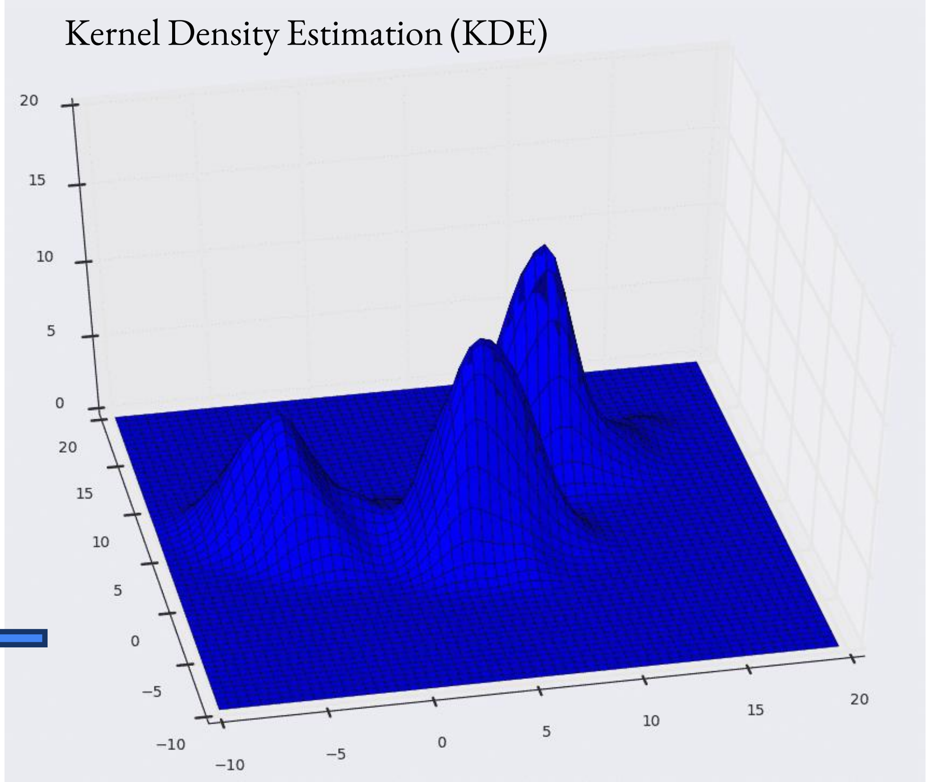

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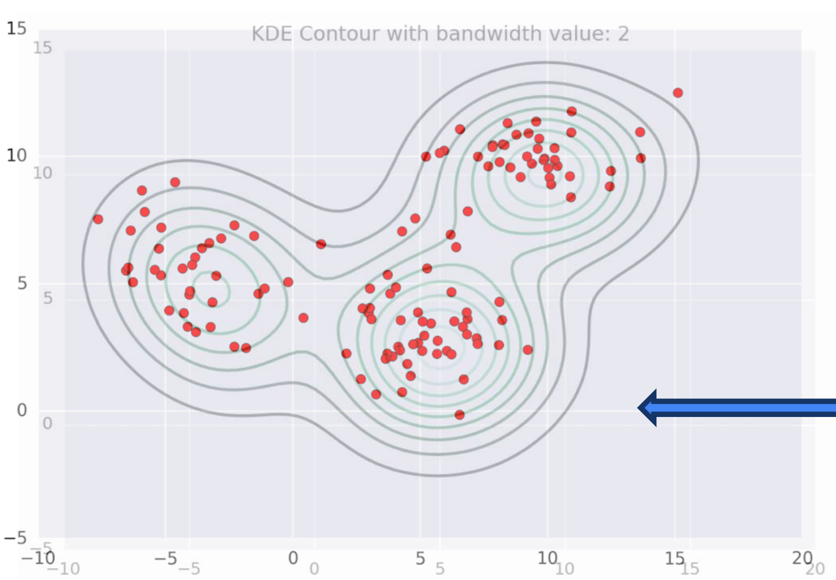
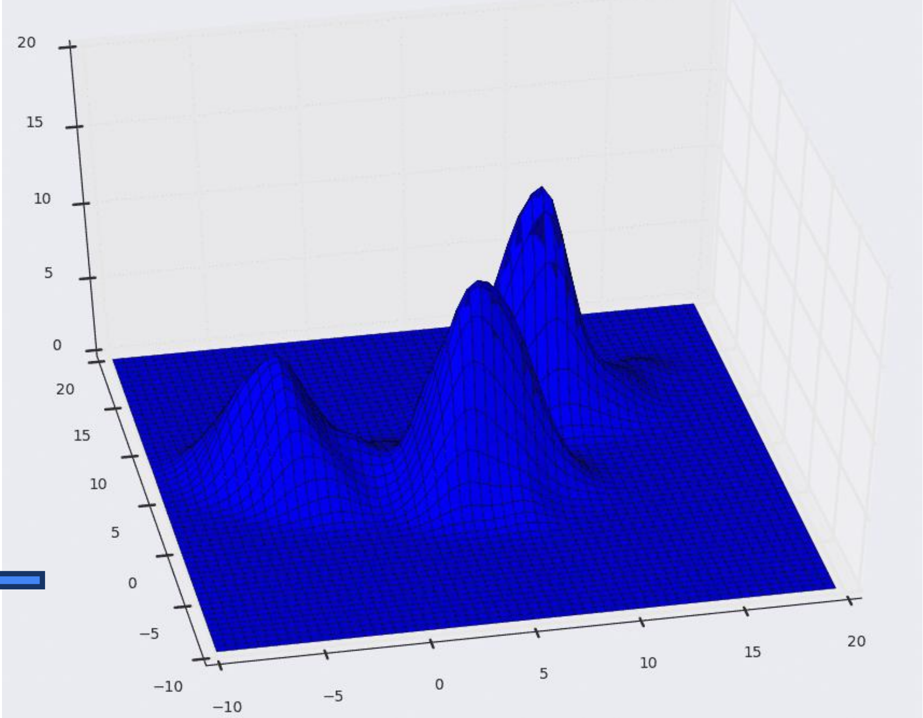


Kernel Density Estimation (KDE)



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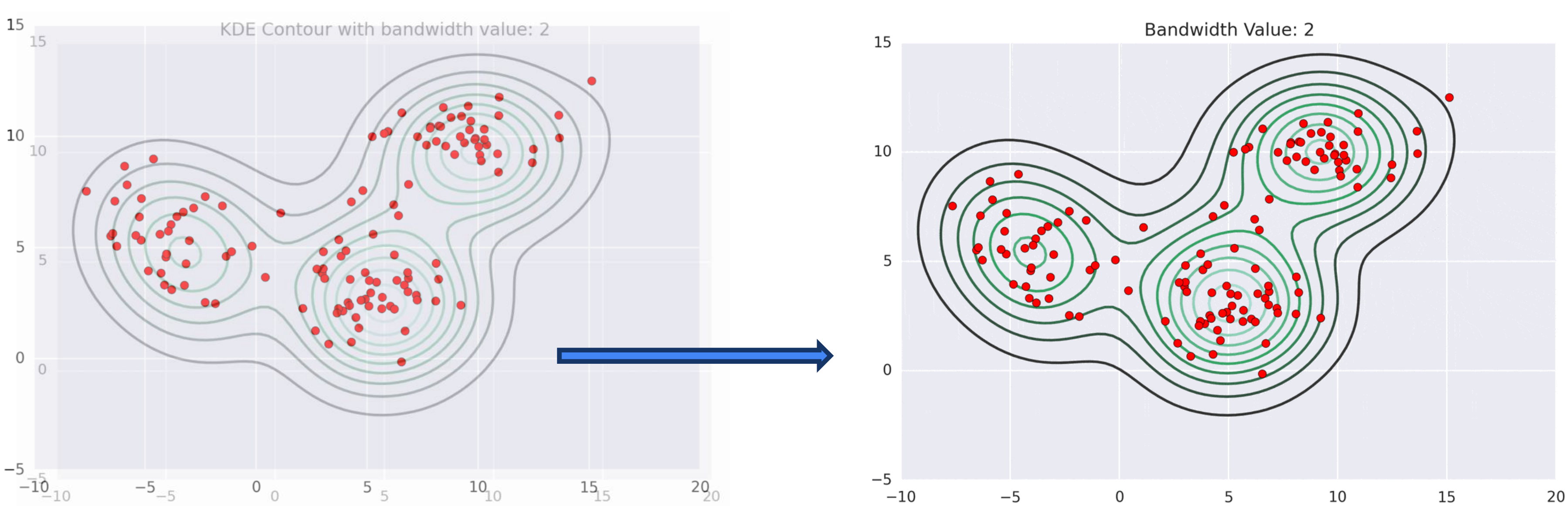

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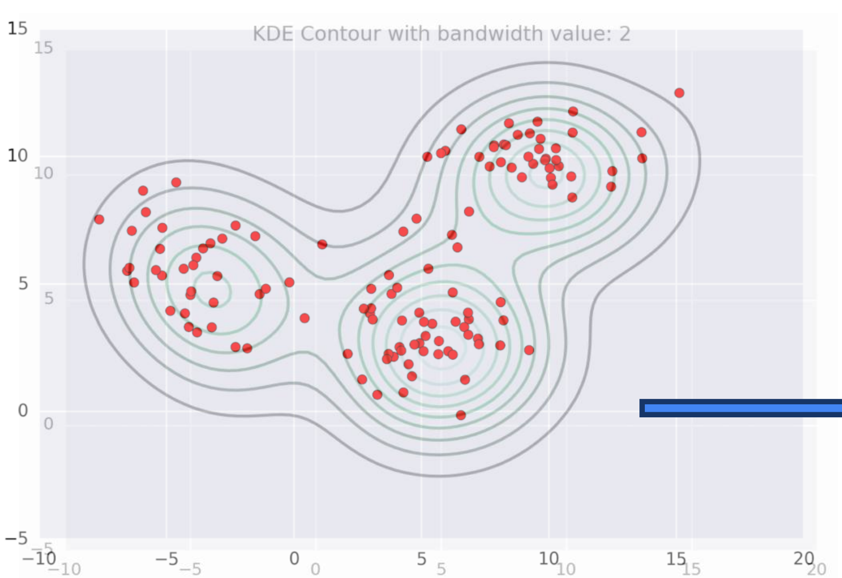
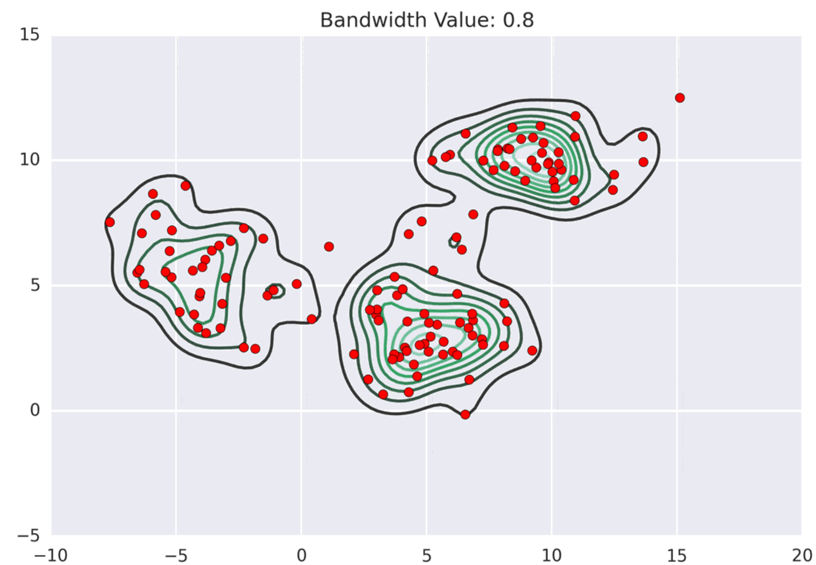
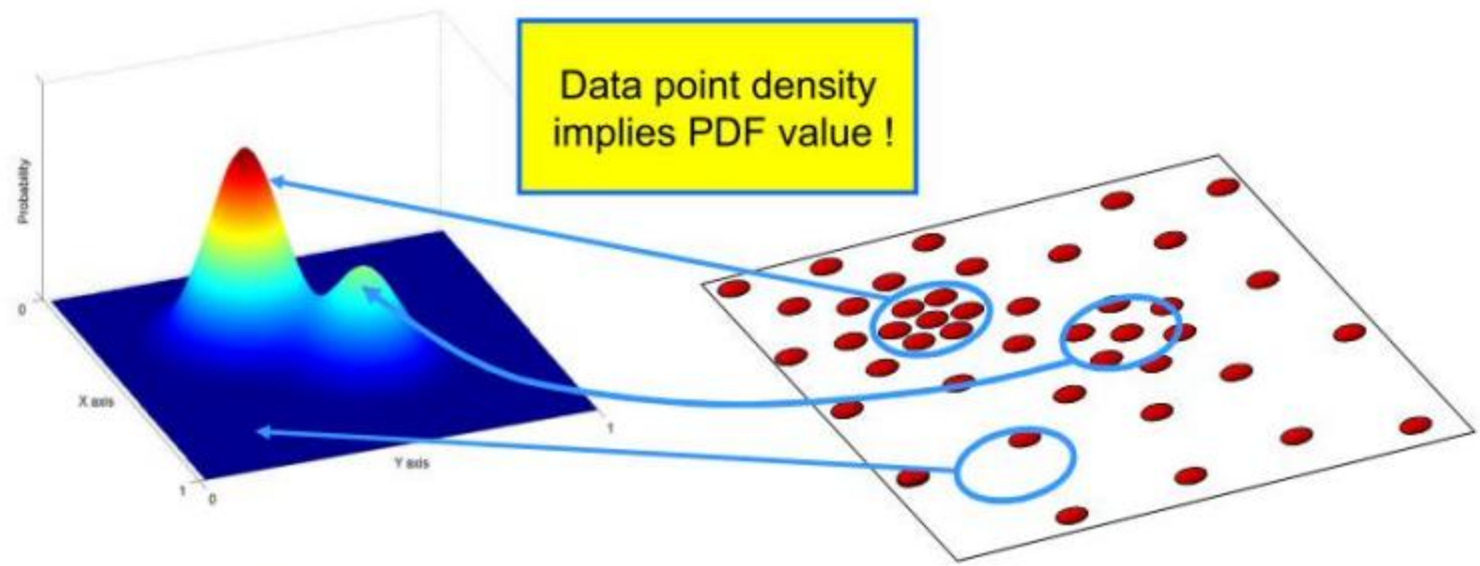




Mean-Shift clustering

Non-Parametric Density Estimation

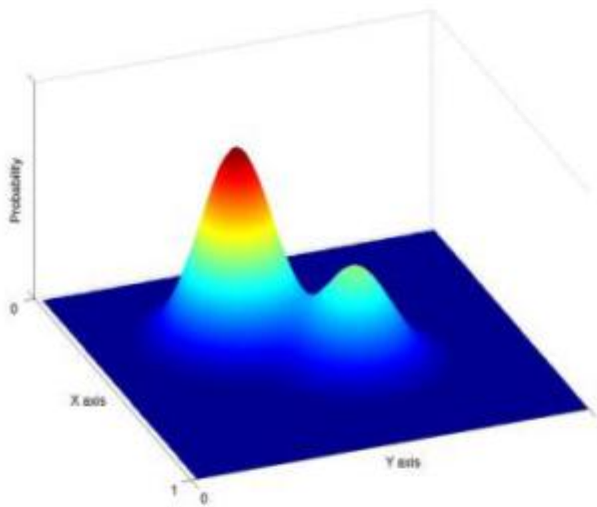
Assumption : The data points are sampled from an underlying PDF



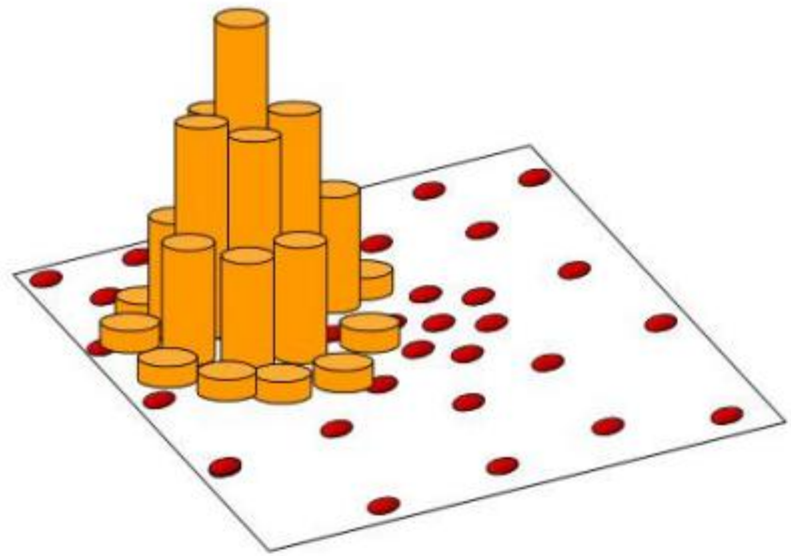
Assumed Underlying PDF **Real Data Samples**

Mean-Shift clustering

Non-Parametric Density Estimation



Assumed Underlying PDF



Real Data Samples

Mean-Shift clustering

mean shift

Consider a dataset $\{\mathbf{x}_n\}_{n=1}^{N} \subset \mathbb{R}^D$ and define a kernel density estimate

$$\hat{f}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^{n} K\Big(\frac{\mathbf{x} - \mathbf{x}_i}{h}\Big).$$

mode seeking

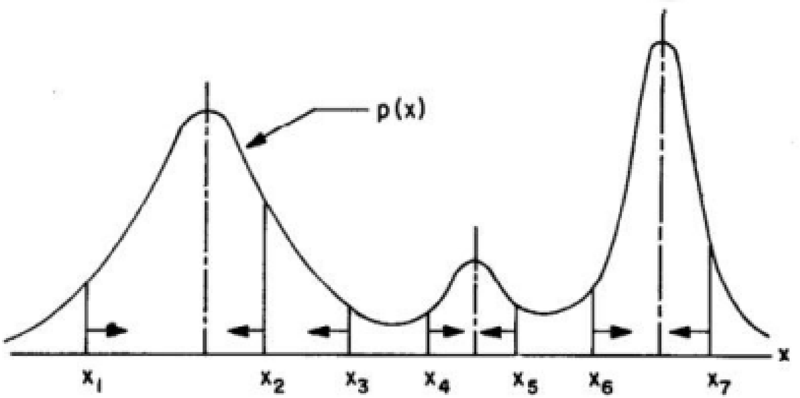


Fig. 1. Gradient mode clustering.

Mean-Shift clustering

mean shift

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$$\hat{f}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{\mathbf{x} - \mathbf{x}_i}{h}\right).$$

Other than estimating the PDF directly, estimating the gradient --

$$\hat{\nabla} f_{h,K}(\mathbf{x}) \equiv \nabla \hat{f}_{h,K}(\mathbf{x}) = \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} (\mathbf{x} - \mathbf{x}_i) k'\left(\left\|\frac{\mathbf{x} - \mathbf{x}_i}{h}\right\|^2\right).$$

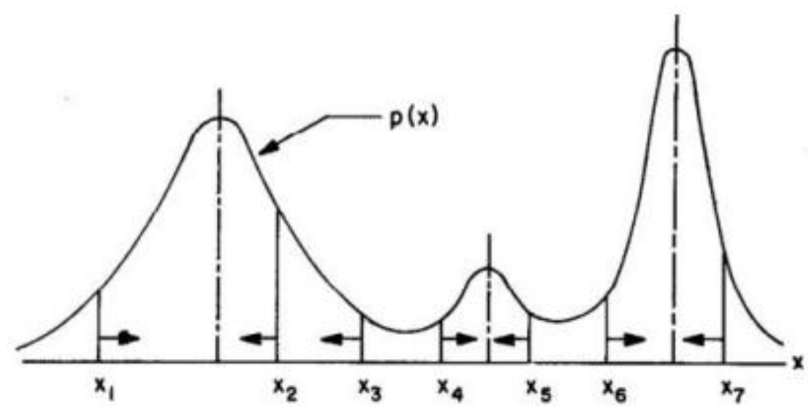

Fig. 1. Gradient mode clustering.

Mean-Shift clustering

mean shift

$$\hat{\nabla} f_{h,K}(\mathbf{x}) \equiv \nabla \hat{f}_{h,K}(\mathbf{x}) = \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} (\mathbf{x} - \mathbf{x}_i) k' \left(\left\| \frac{\mathbf{x} - \mathbf{x}_i}{h} \right\|^2 \right).$$

We define the function

$$g(x) = -k'(x),$$

then

$$\begin{aligned}
&\hat{\nabla} f_{h,K}(\mathbf{x}) \\
&= \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} (\mathbf{x}_i - \mathbf{x}) g\left(\left\| \frac{\mathbf{x} - \mathbf{x}_i}{h} \right\|^2 \right) \\
&= \frac{2c_{k,d}}{nh^{d+2}} \left[\sum_{i=1}^{n} g\left(\left\| \frac{\mathbf{x} - \mathbf{x}_i}{h} \right\|^2 \right) \right] \left[\frac{\sum_{i=1}^{n} \mathbf{x}_i g\left(\left\| \frac{\mathbf{x}-\mathbf{x}_i}{h} \right\|^2 \right)}{\sum_{i=1}^{n} g\left(\left\| \frac{\mathbf{x}-\mathbf{x}_i}{h} \right\|^2 \right)} - \mathbf{x} \right],
\end{aligned}$$

$\sum_{i=1}^{n} g\left(\left\| \frac{\mathbf{x}-\mathbf{x}_i}{h} \right\|^2 \right)$ is assumed to be a positive number.

Mean-Shift clustering

mean shift: iteratively updating by shifting the data by such an amount

The second term is the *mean shift*

$$\mathbf{m}_{h,G}(\mathbf{x}) = \frac{\sum_{i=1}^{n} \mathbf{x}_i g\left(\left\|\frac{\mathbf{x}-\mathbf{x}_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} g\left(\left\|\frac{\mathbf{x}-\mathbf{x}_i}{h}\right\|^2\right)} - \mathbf{x},$$

$$\mathbf{Y} = \mathbf{X} + \eta \mathbf{M}$$

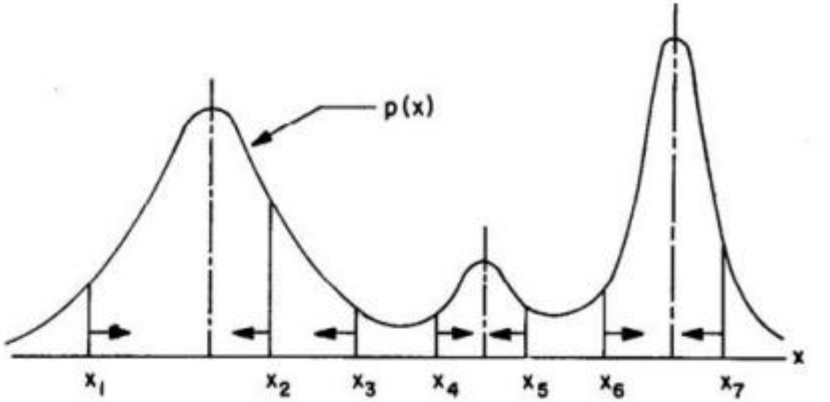

Fig. 1. Gradient mode clustering.

Mean-Shift clustering

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Gaussian blurring mean-shift (GBMS) algorithm

the new iterate is the data average under the posterior probabilities given the current iterate:

repeat — Iteration loop

  for $m \in \{1, \ldots, N\}$ — For each data point

    $\forall n\colon p(n|\mathbf{x}_m) \leftarrow \dfrac{e^{-\frac{1}{2}\left\|\frac{\mathbf{x}_m - \mathbf{x}_n}{\sigma}\right\|^2}}{\sum_{n'=1}^{N} e^{-\frac{1}{2}\left\|\frac{\mathbf{x}_m - \mathbf{x}_{n'}}{\sigma}\right\|^2}}$ — Eq. (3a)

    $\mathbf{y}_m \leftarrow \sum_{n=1}^{N} p(n|\mathbf{x}_m)\mathbf{x}_n$ — One GMS step, eq. (3b)

  end

Mean-Shift clustering

Gaussian blurring mean-shift (GBMS) algorithm

It's guaranteed to converge without gradient vanishing or exploding

B. Gaussian blurring mean-shift (GBMS) algorithm

repeat — Iteration loop

  for $m \in \{1, \dots, N\}$ — For each data point

    $\forall n$: $p(n|\mathbf{x}_m) \leftarrow \dfrac{e^{-\frac{1}{2}\left\|\frac{\mathbf{x}_m - \mathbf{x}_n}{\sigma}\right\|^2}}{\sum_{n'=1}^{N} e^{-\frac{1}{2}\left\|\frac{\mathbf{x}_m - \mathbf{x}_{n'}}{\sigma}\right\|^2}}$ — Eq. (3a)

    $\mathbf{y}_m \leftarrow \sum_{n=1}^{N} p(n|\mathbf{x}_m)\mathbf{x}_n$ — One GMS step, eq. (3b)

  end

  $\forall m$: $\mathbf{x}_m \leftarrow \mathbf{y}_m$ — Update whole dataset

until stop — See section 1

connected-components($\{\mathbf{x}_n\}_{n=1}^{N}$,`min_diff`) — Clusters

C. GBMS algorithm in matrix form

repeat — Iteration loop

  $\mathbf{W} = \left(\exp\left(-\frac{1}{2}\left\|\frac{\mathbf{x}_m - \mathbf{x}_n}{\sigma}\right\|^2\right)\right)_{nm}$ — Gaussian affinity matrix

  $\mathbf{D} = \operatorname{diag}\left(\sum_{n=1}^{N} w_{nm}\right)$ — Degree (normalising) matrix

  $\mathbf{X} = \mathbf{XWD}^{-1}$ — Update whole dataset

until stop — See section 1

connected-components($\{\mathbf{x}_n\}_{n=1}^{N}$,`min_diff`) — Clusters